MATRICES

SYNOPSIS

- **Definition :** A rectangular arrangement of numbers (which may be real or complex numbers) in rows and columns, is called a matrix. This arrangement is enclosed by open ( ) or closed [ ] brackets. The numbers are called the elements of the matrix or entries of the matrix.
- **Order of Matrix :** A matrix having 'm' rows and 'n' columns is called a matrix of order '$m \times n$' or simply '$m \times n$' matrix (read as m by n matrix). A matrix A of order '$m \times n$' is usually written in the following manner

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & \dots & a_{1j} & \dots & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & \dots & a_{2j} & \dots & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{i1} & a_{i2} & a_{i3} & \dots & \dots & a_{ij} & \dots & \dots & a_{in} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & \dots & a_{mj} & \dots & \dots & a_{mn} \end{bmatrix}_{m\times n}$$

In a compact form the above matrix is represented by

$A = \left[a_{ij}\right]_{m\times n}, 1 \le i \le m, 1 \le j \le n \,(or)\, A = \left[a_{ij}\right]$

The numbers $a_{11}, a_{12}, \ldots\ldots,$ etc. are known as the elements of the matrix A. The element a_{ij} in the matrix A belongs to i^{th} row and j^{th} column.

Note: A matrix of order '$m \times n$' contains mn elements. Every row of such a matrix contains n elements and every column contains m elements.

Eg : Order of matrix $\begin{bmatrix} 3 & -1 & 5 \\ 6 & 2 & -7 \end{bmatrix}$ is 2×3

- **Types of matrices:**

Row matrix: A matrix is said to be a row matrix if it has only one row and any number of columns.

Eg : $[5\ 0\ 3]$ is a row matrix of order 1×3

- **Column matrix :** A matrix is said to be a column matirx if it has only one column and any number of rows.

Eg : $\begin{bmatrix} 2 \\ 3 \\ -6 \end{bmatrix}$ is a column matrix of order 3×1

- **Singleton matrix :** If in a matrix there is only one element then it is called singleton matrix.

Thus, $A = \left[a_{ij}\right]_{m\times n}$ is a singleton matrix, if m = n = 1

Eg : $[2],[3],[a],[-3]$ are singleton matrices.

- **Null matrix (or) Zero matrix :** If in a matrix all the elements are zeros then it is called a zero matrix and it is generally denoted by O. Thus

$A = \left[a_{ij}\right]_{m\times n}$ is a zero matrix if $a_{ij} = 0 \,\forall$ i and j.

Eg : $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ are all zero matrices, but of different orders.

- **Rectangular matrix :** In a matrix if the number of rows is not equal to the number of columns then the matrix is called a rectangular matrix.

Eg : $\begin{bmatrix} 3 & -2 \\ 5 & 0 \\ 7 & 1 \end{bmatrix}_{3\times2}$

- **Square matrix :** If number of rows and number of columns in a matrix are equal, then it is called a square matrix.

Eg : $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ is a square matrix of order 3 × 3.

- **Principal diagonal of a square matrix :** In a square matrix the diagonal from first element of the first row to the last element of the last row is called the principal diagonal of the square matrix.

Eg : $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

then a_{11}, a_{22}, a_{33} constitutes diagonal of A

- **Diagonal matrix :** In a square matrix if all the elements outside the principal diagonal are zeros, the elements of principal diagonal may or may not be zero, then the matrix is said to be diagonal matrix.

Eg : $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ is a diagonal matrix of order 3×3, which can be denoted by diag $[2, 0, 4]$

Note: If A = diag($d_1, d_2, d_3 ... d_n$) then

$A^n = diag\left(d_1^{\,n}, d_2^{\,n}, d_3^{\,n} .....d_n^{\,n}\right)$

- **Identity matrix or Unit matrix :** A square matrix in which each element in the principal diagonal is '1' and rest are all zeros is called an identity matrix or unit matrix. Thus, the square matrix $A = \left[a_{ij}\right]$ is an identity matrix,

if $a_{ij} = \begin{cases} 1, \text{ if } i = j \\ 0, \text{ if } i \neq j \end{cases}$

We denote the identity matrix of order n by I_n.

Eg : $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ are identity matrices of order 2 and 3 respectively.

- **Scalar matrix :** A square matrix whose all non diagonal elements are zeros and diagonal elements are equal is called a scalar matrix. Thus, if $A = \left[a_{ij}\right]$ is a square matrix and

$a_{ij} = \begin{cases} \alpha, \text{if } i = j \\ 0, \text{ if } i \neq j \end{cases}$, then A is a scalar matrix.

Eg : $\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}, \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

Unit matrix and null square matrices are also scalar matrices.

- **Triangular matrix :** A square matrix $\left[a_{ij}\right]$ is said to be triangular matrix if each element above or below the principal diagonal is zero. It is of two types

Upper triangular matrix: A square matrix $\left[a_{ij}\right]$ is called the upper triangular matrix, if $a_{ij} = 0$ when $i > j$.

Eg : $\begin{bmatrix} 3 & 1 & 2 \\ 0 & 4 & 3 \\ 0 & 0 & 6 \end{bmatrix}$ is an upper triangular matrix of order 3 × 3.

Lower triangular matrix: A square matrix $\left[a_{ij}\right]$ is called the lower triangular matrix, if $a_{ij} = 0$ when $i < j$.

Eg : $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 2 \end{bmatrix}$ is a lower triangular matrix of order 3 × 3

- **Trace of a matrix :** The sum of diagonal elements of a square matrix A is called the trace of matrix A, which is denoted by tr (A). Trace is also called as spur.

i.e., tr $(A) = \sum_{i=1}^{n} a_{ii} = a_{11} + a_{22} + ...a_{nn}$

Properties of trace of a matrix :

Let $A = \left[a_{ij}\right]_{n\times n}$ and $B = \left[b_{ij}\right]_{n\times n}$ and λ be a scalar i) $tr(\lambda A) = \lambda tr(A)$

ii) $tr(A \pm B) = tr(A) \pm tr(B)$

iii) $tr(AB) = tr(BA)$

iv) $tr(I_n) = n$ v) $tr(AB) \neq tr(A).tr(B)$

vi) If A,B,C are square matrices of order n, then Tr (ABC) = Tr (BCA) = Tr (CAB)

- **Equality of matrices :** Two matrices A and B are said to be equal, if
  i) A and B are of the same type (order)
  ii) The corresponding elements of A and B are equal.

Eg : $A=\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$ and $B=\begin{bmatrix} 2 & 0 & 1 \\ -1 & -3 & 5 \end{bmatrix}$

are equal iff a = 2, b = 0, c = 1, d = -1, e = -3, f = 5

- **Addition of matrices :** If A and B are two matrices of the same type, then their sum denoted by A+B is defined to be the matrix of the same type and is obtained by adding the corresponding elements of A and B.

i.e., If $A=\left[a_{ij}\right]_{m\times n}$ and $B=\left[b_{ij}\right]_{m\times n}$ then

$A+B=[C_{ij}]_{m\times n}$ where $C_{ij}=a_{ij}+b_{ij}$

- **Properties :**
  i) A+B = B+A (commutative law)
  ii) (A+B)+C = A+(B+C) (Associative law)
  iii) A+O=O+A = A where O is zero matrix which is called additive identity.
  iv) $A+(-A)=O=(-A)+A$, where $(-A)$ is obtained by changing the sign of every element of A, which is additive inverse of the matrix A.

- **Difference of two matrices :** If A and B are two matrices of the same type then A-B is defined as A+(-B)

Note: If two matrices A and B are of different orders, we can not define A+B and A-B.

- **Multiplication of a matrix by a scalar:** Let $A=\left[a_{ij}\right]_{m\times n}$ be a matrix and k be a number, then the matrix which is obtained by multiplying every element of A by k is called scalar multiplication of A by k and it is denoted by kA.

Thus, if $A=\left[a_{ij}\right]_{m\times n}$,

then $kA=Ak=\left[ka_{ij}\right]_{m\times n}$.

- **Properties of scalar multiplication :** If A, B are matrices of the same order and λ,μ are any two scalars then
  i) $\lambda(A+B)=\lambda A+\lambda B$
  ii) $(\lambda+\mu)A=\lambda A+\mu A$
  iii) $\lambda(\mu A)=(\lambda\mu A)=\mu(\lambda A)$
  iv) $(-\lambda A)=-(\lambda A)=\lambda(-A)$

- **Multiplication of matrices :** Two matrices A and B are conformable for the product AB if the number of columns in A is same as the number of rows in B thus, if

$A=\left[a_{ik}\right]_{m\times p}$, $B=\left[b_{kj}\right]_{p\times n}$ then

$AB=\left[C_{ij}\right]_{m\times n}$ where $C_{ij}=\sum_{k=1}^{p}a_{ik}b_{kj}$

***W.E-1:* A matrix X has (a+b) rows and (a+2) columns, while the matrix Y has (b+1) rows and (a+3) columns. Both matrices XY and YX exist. Find a and b.**

Sol. Type of X is (a+b) $\times$ (a+2)
Type of Y is (b+1) $\times$ (a+3)
Since both XY, YX exist
$\therefore$ a+2=b+1 and a+b=a+3 $\therefore$ a=2, b=3
order of X is 5×4, order of Y is 4×5
$\therefore$ XY $\neq$ YX
($\because$ order of XY is 5×5 and YX is 4×4)

- **Properties of matrix multiplication :** If A,B and C are three matrices such that their product is defined, then
  i) $AB\neq BA$,(Generally not commutative)
  ii) $(AB)C=A(BC)$=ABC, (Associative law)
  iii) $IA=A=AI$, where I is identitiy matrix for matrix multiplication.
  iv) $A(B+C)=AB+AC$, (Distributive law)
  v) (A + B).C = AC + BC, (Distributive law)
  vi) If $AB=O$, then either A or B need not be equal to O.
  vii) If AB = AC then B need not be equal to C even if $A\neq O$.

- Remember that if A and B are two square matrices of the same order, then

i) $(A+B)^2 = A^2 + B^2 + AB + BA$

ii) $(A-B)^2 = A^2 + B^2 - AB - BA$

iii) $A^m A^n = A^{m+n}$ iv) $(A^m)^n = A^{mn} = (A^n)^m$

v) $I^n = I \ \forall n \in N$

vi) If a square matrix, which is commutative with every square matrix of the same order for multiplication then it is necessarily a scalar matrix.

➢ **Transpose of a matrix :** The matrix obtained from a given matrix A by changing its rows into columns or columns into rows is called transpose of matrix A and is denoted by A^T or A'

From the definition it is obvious that if order of A is $m\times n$, then order of A^T is $n\times m$

➢ **Properties of transpose of a matrix :**

Let A and B be two matrices then, (i) $(A^T)^T = A$

ii) $(A\pm B)^T = A^T \pm B^T$,A and B being of the same order

iii) $(kA)^T = kA^T$, k be any scalar

iv) $(AB)^T = B^T A^T$, A and B being conformable for the product AB

v) $(A_1A_2A_3....A_{n-1}A_n)^T = A_n^T A_{n-1}^T \ ....A_3^T A_2^T A_1^T$

vi) A=B $\Leftrightarrow A^T = B^T$

➢ **Symmetric matrix:** A square matrix $A=[a_{ij}]$ is called symmetric matrix if $a_{ij} = a_{ji}$ $\forall\ i,j$ i.e., $A^T = A$

Eg : $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ is a symmetric matrix

➢ **Skew-Symmetric matrix :** A square matrix $A=[a_{ij}]$ is called skew -symmetric matrix if $a_{ij} = -a_{ji}$ $\forall$ i, j i.e., $A^T = -A$.

Eg : $\begin{bmatrix} 0 & h & g \\ -h & 0 & f \\ -g & -f & 0 \end{bmatrix}$

Note: i) All principal diagonal elements of a skew-symmetric matrix are always zeros

ii) Trace of a skew-symmetric matrix is zero

➢ **Properties of symmetric and Skew-symmetric matrices :**

i) If A is a square matrix, then $A+A^T, AA^T, A^TA$ are symmetric matrices

ii) If A is square matrix then $A-A^T$ is a skew-symmetric matrix.

iii) If A is a symmetric matrix, then $-A, KA, A^T, A^n, A^{-1}, B^TAB$ are also symmetric matrices, where $n\in N$, $K \in R$ and B is a square matrix of order that of A

iv) If A is a skew-symmetric matrix, then

a) A^{2n} is symmetric matrix for $n \in N$

b) A^{2n+1} is a skew-symmetric matrix for $n\in N$

c) kA is also skew-symmetric matrix, where $k \in R$

v) If A, B are two symmetric matrices, then

a) $A\pm B, AB+BA$ are also symmetric matrices,

b) AB-BA is a skew-symmetric matrix,

c) AB is a symmetric matrix, when AB=BA

vi) If A, B two skew-symmetric matrices, then

a) $A\pm B$,AB-BA are skew-symmetric matrices

b) AB+BA is a symmetric matrix.

vii) a) If A is a skew-symmetric matrix and B is a square matrix of order that of A then B^TAB is also skew-symmetric matrix.

b) If A is a skew-symmetric matrix and C is a column matrix, then C^T AC is a zero matrix.

viii) Every square matrix A can be uniquely expressed as sum of a symmetric and skew-symmetric matrices of same order

ix) If A,B are symmetric matrices of same order and $X = AB+BA$, $y = AB-BA$ then $XY+YX=0$

i.e., $A = \left[\frac{1}{2}(A+A^T)\right] + \left[\frac{1}{2}(A-A^T)\right]$.

Note: If a matrix A is both symmetric and skew-symmetric then A is null matrix.

➢ **Orthogonal matrix:** A square matrix A is called orthogonal if $AA^T = I = A^T A$ i.e.,

$A^{-1} = A^T$

W.E 2:** If* $A = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ ***then A is__

Sol: $A^T = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$

$$A.A^T = \frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}\frac{1}{3}\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

$$= \frac{1}{9}\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = I$$

Similarly $A^T A = I$

$\therefore$ A is Orthogonal matrix

➢ **Properties of Orthogonal matrix :**

i) Every orthogonal matrix is non -singular.
ii) Every orthogonal matrix is invertible.
iii) If A is orthogonal, then A^T *and* A^{-1} are also orthogonal.
iv) If A and B are orthogonal matrices of same order then AB and BA are also orthogonal.
v) The sum of the squares of elements of any row or column of an orthogonal matrix is 1.
vi) The sum of the products of the corresponding elements of any two rows or columns is 0
vii) If A is an orthogonal matrix and B = AP (P-non-singular matrix) then PB^{-1} is also orthogonal matrix.

➢ **Special types of matrices :**

i) Idempotent matrix: A square matrix A is called an idempotent matrix if $A^2 = A$

Eg : A = $\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is an idempotent matrix.

Properties :

i) If A, B are two idempotent matrices and AB = BA = O then A+B is idempotent matrix.
ii) If A is idempotent matrix then I-A is also idempotent
iii) Every non-singular idempotent matrix is always unit matrix
iv) If AB=A , BA=B then A and B are idempotent matrices and $A^n + B^n = A + B$

ii) Involutory matrix: A square matrix A is called an involutory matrix if $A^2 = I$

Eg : $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is an involutory matrix.

iii) Nilpotent matrix: A square matrix A is called a nilpotent matrix if there exists atleast one $p \in N$ such that $A^p = O$, where the least value p is called index of the nilpotent matrix A

Eg : **A=** $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is a nilpotent matrix of index 3.

Note: Trace of a nilpotent matrix is zero.

iv) Periodic matrix: A matrix A is called a periodic matrix if $A^{k+1} = A$ where k is a positive integer. The least value of k is said to be period of A

v) Conjugate of a matrix: The matrix obtained from any given matrix A containing complex numbers as its elements, on replacing its elements by the corresponding conjugate complex numbers is called conjugate of A and is denoted by $\overline{A}$.

Eg : A = $\begin{bmatrix} 1+2i & 2-3i & 3+4i \\ 4-5i & 5+6i & 6-7i \\ 8 & 7+8i & 7 \end{bmatrix}$ then

$$\bar{A} = \begin{bmatrix} 1-2i & 2+3i & 3-4i \\ 4+5i & 5-6i & 6+7i \\ 8 & 7-8i & 7 \end{bmatrix}$$

vi) Transposed conjugate of a matrix: The transpose of the conjugate of a matrix A is called transposed conjugate of A and is denoted by A^{θ} or A^{*}. The conjugate of the transpose of A is the same as the transpose of the conjugate of A

i.e., $\left(\overline{A^T}\right) = \left(\overline{A}\right)^T = A^{\theta}$.

Eg : $A = \begin{bmatrix} 1+2i & 2-3i & 3+4i \\ 4-5i & 5+6i & 6-7i \\ 8 & 7+8i & 7 \end{bmatrix}$

then $A^{\theta} = \begin{bmatrix} 1-2i & 4+5i & 8 \\ 2+3i & 5-6i & 7-8i \\ 3-4i & 6+7i & 7 \end{bmatrix}$

vii) Hermitian matrix: A square matrix 'A' is said to be Hermitian matrix if $A^{\theta} = A$

Eg : $\begin{bmatrix} 3 & 3-4i & 5+2i \\ 3+4i & 5 & -2+i \\ 5-2i & -2-i & 2 \end{bmatrix}$ is Hermitian matrix.

Note: All the principal diagonal elements of herimitian matrix are real.

viii) Skew Hermitian matrix : A square matrix A is said to be a skew-Hermitian if

$A^{\theta} = -A$

Eg : $\begin{bmatrix} 3i & -3+2i & -1-i \\ 3+2i & -2i & -2-4i \\ 1-i & 2-4i & 0 \end{bmatrix}$

is a skew-hermitian matrix.

Note: All the principal diagonal elements of skew-hermitian matrix are either zero or purely imaginary.

ix) Unitary matrix: A square matrix A is said to be unitary, if $A^{\theta}A = I = AA^{\theta}$

The determinant of unitary matrix is one and it is non-singular.

➢ **Determinant of a square matrix :**

The determinant of the square matrix

$A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$ is the unique number $a_1b_2 - a_2b_1$ and is denoted by

$$\text{Det } A = |A| = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1b_2 - a_2b_1$$

➢ **Minor of an element in a square matrix:** Let $A = [a_{ij}]$ be a square matrix. The minor of an element a_{ij} in A is the determinant of the square matrix that remains after deleting the i^{th} row and j^{th} column of A. It is denoted by M_{ij}.

➢ **Cofactor of an element of a square matrix :** The cofactor of an element in the i^{th} row and the j^{th} column of a matrix is defined as its minor multiplied by $(-1)^{i+j}$. If A_{ij} is the cofactor of a_{ij}, then $A_{ij} = (-1)^{i+j}M_{ij}$

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

i) Minor of a_{23} is $(M_{23}) = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$

$= a_{11}a_{32} - a_{31}a_{12}$

ii) Cofactor of a_{31} is

$$A_{31} = (-1)^{3+1}M_{31} = +\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$$

iii) The cofactors of $a_{11}, a_{12}, a_{13}.....$ are denoted by $A_{11}, A_{12}, A_{13},.......$

iv) For the matrix $\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, the cofactor matrix is given by $\begin{bmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{bmatrix}$

Eg : $B_2 = (-1)^{2+2}\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix} = (a_1c_3 - a_3c_1)$

➢ Determinant of Third Order Matrix

The determinant of a square matrix is equal to the sum of the products of the elements of a row (or column) with their corresponding cofactors

If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow \Delta = a_1 A_1 + a_2 A_2 + a_3 A_3$

$= b_1 B_1 + b_2 B_2 + b_3 B_3$
$= c_1 C_1 + c_2 C_2 + c_3 C_3$

Properties of determinants:

i) The determinant of a square matrix changes its sign when any two rows (or columns) are interchanged

ii) If two rows (columns) of a square matrix are identical or proportional then the value of the determinant is zero.

iii) If all the elements of a row (column) of a square matrix are multiplied by a number K then the determinant of the resulting matrix is equal to K times the determinant of the original matrix.

iv) If each element of a row (column) of a square matrix is the sum of two terms then its determinant can be expressed as the sum of two determinants of two square matrices of the same order.

$$\begin{vmatrix} a_1+x & a_2 & a_3 \\ b_1+y & b_2 & b_3 \\ c_1+z & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} x & a_2 & a_3 \\ y & b_2 & b_3 \\ z & c_2 & c_3 \end{vmatrix}$$

v) If the elements of a row (column) of a square matrix are added with K times the corresponding elements of any other row (column) then the value of the determinant of the resulting matrix is unaltered

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \begin{vmatrix} x_1+ky_1 & y_1 & z_1 \\ x_2+ky_2 & y_2 & z_2 \\ x_3+ky_3 & y_3 & z_3 \end{vmatrix}$$

vi) The sum of the products of the elements of any row (column) of a squre matrix with the cofactors of the corresponding elements of any other row (column) is zero.

$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow a_1 A_2 + b_1 B_2 + c_1 C_2 = 0$

$a_1 A_3 + b_1 B_3 + c_1 C_3 = 0$

$a_2 A_1 + b_2 B_1 + c_2 C_1 = 0$

vii) If the elements of a square matrix are Polynomials in x and two rows (columns) become identical when $x = a$ then $x-a$ is a factor of its determinant and

if three rows are identical then $(x-a)^2$ is a factor

Note: i) $|A| = |A^T|, |AB| = |A||B| = |B||A|$

ii) $|KA| = K^n |A|$, $n =$ order of A.

iii) If $\Delta = |a_{ij}|$ is a determinant of order n, then the value of the determinant $[A_{ij}]$, where A_{ij} is the cofactor of a_{ij} is, Δ^{n-1}.

iv) Determinant of nilpotent matrix is 0

v) Determinant of an orthogonal matrix = 1 or -1

vi) Determinant of a Skew - symmetric matrix of odd order is 0.

vii) Determinant of Hermitian matrix is purely real.

viii) Determinant of triangular matrix is zero

W.E-3: In a square matrix, the elements of a column are 2, 5k+1, 3 and the cofactors of another column are 1-5k, 2, 4k-2. Then find k

Sol. The sum of the products of the elements of any column of a square matrix with the cofactors of the corresponding elements of any other column is zero.

$\therefore\ 2(1-5k) + (5k+1)2 + 3(4k-2) = 0$

$12k - 2 = 0 \Rightarrow k = \frac{1}{6}$

W.E-4 : If α, β, γ are roots of $x^3 + px^2 + q = 0$,

where $q \neq 0$ then $\begin{vmatrix} \frac{1}{\alpha} & \frac{1}{\beta} & \frac{1}{\gamma} \\ \frac{1}{\beta} & \frac{1}{\gamma} & \frac{1}{\alpha} \\ \frac{1}{\gamma} & \frac{1}{\alpha} & \frac{1}{\beta} \end{vmatrix} = -$

Sol. we have $\alpha\beta + \beta\gamma + \gamma\alpha = 0 \Rightarrow \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 0$

Let $\Delta = \begin{vmatrix} \frac{1}{\alpha} & \frac{1}{\beta} & \frac{1}{\gamma} \\ \frac{1}{\beta} & \frac{1}{\gamma} & \frac{1}{\alpha} \\ \frac{1}{\gamma} & \frac{1}{\alpha} & \frac{1}{\beta} \end{vmatrix}$ $C_1 \to C_1 + C_2 + C_3$

$$= \begin{vmatrix} \frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma} & \frac{1}{\beta} & \frac{1}{\gamma} \\ \frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma} & \frac{1}{\gamma} & \frac{1}{\alpha} \\ \frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma} & \frac{1}{\alpha} & \frac{1}{\beta} \end{vmatrix} = \begin{vmatrix} 0 & \frac{1}{\beta} & \frac{1}{\gamma} \\ 0 & \frac{1}{\gamma} & \frac{1}{\alpha} \\ 0 & \frac{1}{\alpha} & \frac{1}{\beta} \end{vmatrix} = 0$$

W.E-5: $\begin{vmatrix} 1 & 1+p & 1+p+q \\ 2 & 3+2p & 4+3p+2q \\ 3 & 6+3p & 10+6p+3q \end{vmatrix} =$

Sol. $\Delta = \begin{vmatrix} 1 & 1+p & 1+p+q \\ 2 & 3+2p & 4+3p+2q \\ 3 & 6+3p & 10+6p+3q \end{vmatrix}$

$c_2 \to c_2 - pc_1$ *and* $c_3 \to c_3 - qc_1$

$$\Delta = \begin{vmatrix} 1 & 1 & 1+q \\ 2 & 3 & 4+3p \\ 3 & 6 & 10+6p \end{vmatrix}_{C_3 \to C_3 - PC_2}$$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 6 & 10 \end{vmatrix} = 1$$

W.E- 6. ***If*** $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & x(x+1) \\ 3x(x-1) & x(x-1)(x-2) & x(x^2-1) \end{vmatrix}$

then $f(2015) =$ ***[EAM-2012]***

Sol. Taking x common from C_2, $x+1$ from C_3 and (x-1) from R_3, we get

$$f(x) = x(x+1)(x-1)\begin{vmatrix} 1 & 1 & 1 \\ 2x & x-1 & x \\ 3x & x-2 & x \end{vmatrix}$$

Applying $C_1 \to C_1 - C_3,\ C_2 \to C_2 - C_3$, we get

$$f(x) = x(x^2-1)\begin{vmatrix} 0 & 0 & 1 \\ x & -1 & x \\ 2x & -2 & x \end{vmatrix}$$

=0 [$\because C_1$ and C_2 are proportional]

Thus f(2015) = 0

➤ Product of Determinants of the same Order :

Let $\Delta_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ *and* $\Delta_2 = \begin{vmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{vmatrix}$

Then row by row multiplication of Δ_1 *and* Δ_2 is given by

$$\Delta_1\Delta_2 = \begin{vmatrix} a_1\alpha_1 + b_1\beta_1 + c_1\gamma_1 & a_1\alpha_2 + b_1\beta_2 + c_1\gamma_2 & a_1\alpha_3 + b_1\beta_3 + c_1\gamma_3 \\ a_2\alpha_1 + b_2\beta_1 + c_2\gamma_1 & a_2\alpha_2 + b_2\beta_2 + c_2\gamma_2 & a_2\alpha_3 + b_2\beta_3 + c_2\gamma_3 \\ a_3\alpha_1 + b_3\beta_1 + c_3\gamma_1 & a_3\alpha_2 + b_3\beta_2 + c_3\gamma_2 & a_3\alpha_3 + b_3\beta_3 + c_3\gamma_3 \end{vmatrix}$$

Note: Multiplication can also be performed row by column; column by row or column by column as required in the problem.

➤ Derivative of Determinants :

Let $\Delta(x)=\begin{vmatrix} f_1(x) & g_1(x) \\ f_2(x) & g_2(x) \end{vmatrix}$,

where $f_1(x), f_2(x), g_1(x)$ and $g_2(x)$ are the functions of x. then,

$$\Delta'(x)=\begin{vmatrix} f_1'(x) & g_1'(x) \\ f_2(x) & g_2(x) \end{vmatrix}+\begin{vmatrix} f_1(x) & g_1(x) \\ f_2'(x) & g_2'(x) \end{vmatrix}$$

Also

$$\Delta'(x)=\begin{vmatrix} f_1'(x) & g_1(x) \\ f_2'(x) & g_2(x) \end{vmatrix}+\begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$$

Thus, to differentiate a determinant, we differentiate one row (or column) at a time, keeping others unchanged.

W.E-7: If $\Delta_1=\begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$, $\Delta_2=\begin{vmatrix} x & b \\ a & x \end{vmatrix}$ ***are the two given determinants then*** $\frac{d}{dx}(\Delta_1)=$

Sol: Given $\Delta_1=\begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ $\left(\because \frac{d}{dx}(constant)=0,\ \frac{d}{dx}(x)=1\right)$

$$\frac{d}{dx}(\Delta_1)=\begin{vmatrix} 1 & 0 & 0 \\ a & x & b \\ a & a & x \end{vmatrix}+\begin{vmatrix} x & b & b \\ 0 & 1 & 0 \\ a & a & x \end{vmatrix}+\begin{vmatrix} x & b & b \\ a & x & b \\ 0 & 0 & 1 \end{vmatrix}$$

$$=\begin{vmatrix} x & b \\ a & x \end{vmatrix}+\begin{vmatrix} x & b \\ a & x \end{vmatrix}+\begin{vmatrix} x & b \\ a & x \end{vmatrix}=3\Delta_2$$

➢ Integration of Determinants :

If $\Delta(x)=\begin{vmatrix} f(x) & g(x) \\ \lambda_1 & \lambda_2 \end{vmatrix}$, then

$$\int_a^b \Delta(x)\,dx=\begin{vmatrix} \int_a^b f(x)\,dx & \int_a^b g(x)\,dx \\ \lambda_1 & \lambda_2 \end{vmatrix}$$

Here, $f(x)$ and $g(x)$ are functions of x and λ_1, λ_2 are constants.

Note: This formula is only applicable, if there is a variable only in one row or column, otherwise expand the determinant and then integrate.

➢ Singular and non-singular matrix :

If the determinant of a square matrix is zero then it is called a singular matrix otherwise non-singular matrix.

Note: i) A is singular $\Leftrightarrow A^T$ is singular

A is non-singular $\Leftrightarrow A^T$ is non-singular

ii) If A and B are non-singular matrices of the same type, then AB is non-singular of the same type.

iii) If product of two non-zero square matrices is a zero matrix, then both of them must be singular matrices.

➢ Adjoint matrix of a square matrix :

If the elements of a square matrix are replaced by corresponding co-factors then the transpose of the resulting matrix is called the adjoint of the matrix. Adjoint matrix of A is denoted by Adj A

If $P=\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ then $\text{Adj}P=\begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix}$

Where A_1, B_1, C_1..... are the co-factors of a_1, b_1, c_1.....

➢ Properties of adjoint matrix :

If A, B are square matrices of order n and I_n is corresponding unit matrix, then

(i) $A(adjA)=|A|I_n=(adjA)A$

(Thus A $adj(A)$ is always a scalar matrix)

ii) $|adjA|=|A|^{n-1}$

iii) $adj(adjA)=|A|^{n-2}A; |A|\neq 0$

iv) $|adj(adjA)|=|A|^{(n-1)^2}$

v) $|adj(adj(adjA)......r\ times|=|A|^{(n-1)^r}$

vi) $adj(A^T)=(adjA)^T$

vii) $adj(AB)=(adjB)(adjA)$

viii) $adj\left(A^m\right) = \left(adjA\right)^m, m \in N$

ix) $adj\left(kA\right) = k^{n-1}\left(adjA\right), k \in R$

x) $adj\left(I_n\right) = I_n$ xi) $adj\left(O\right) = O$

xii) A is symmetric matrix $\Rightarrow$ adj A is also symmetrix matrix.

xiii) A is diagonal matrix $\Rightarrow$ adj A is also diagonal matrix.

xiv) A is triangular matrix $\Rightarrow adjA$ is also triangular matrix.

xv) A is singular $\Rightarrow |adjA| = 0$

- **Inverse of a matrix :** Let A be a non-singular square matrix of order n, if there exist a square matrix B of the same order such that $AB = BA = I_n$ then B is called the inverse of A and we write it as A^{-1}.

The inverse of A given by $A^{-1} = \dfrac{adj\ A}{\det A}$

A matrix is said to be invertible, if it possesses inverse.

- **Properties of inverse matrix :** If A and B are invertible matrices of the same order, then

i) $\left(A^{-1}\right)^{-1} = A$ ii) $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$

iii) $\left(AB\right)^{-1} = B^{-1}A^{-1}$

iv) $\left(A^k\right)^{-1} = \left(A^{-1}\right)^k, k \in N$

v) $adj\left(A^{-1}\right) = \left(adjA\right)^{-1}$ vi) $\left|A^{-1}\right| = \dfrac{1}{|A|} = |A|^{-1}$

vii) If A= diag($a_1 a_2 ... a_n$)

then $A^{-1} = diag\left(a_1^{-1} a_2^{-1} .... a_n^{-1}\right)$

viii) If A is symmetric matrix then A^{-1} is also symmetric matrix.

ix) The inverse of a skew symmetric matrix of odd order does not exist.

x) A is a non singular scalar matrix $\Rightarrow A^{-1}$ is also a scalar matrix.

xi) A is triangular matrix, $|A| \neq 0 \Rightarrow A^{-1}$ is also triangular matrix.

xii) If A,B are symmetric matrices and commute then $A^{-1}B$, AB^{-1}, $A^{-1}B^{-1}$ are also symmetric matrices

Cancellation law with respect to multiplication : If A is a non -singular matrix i.e., if $|A| \neq 0$, then A^{-1} exist and

$$AB = AC \Rightarrow A^{-1}\left(AB\right) = A^{-1}\left(AC\right)$$

$$\Rightarrow \left(A^{-1}A\right)B = \left(A^{-1}A\right)C \Rightarrow IB = IC \Rightarrow B = C$$

$\therefore AB = AC \Rightarrow B = C \Leftrightarrow |A| \neq 0$.

- **Elementary Transformations :** Any one of the following operation on a matrix is called an elementary transformation.

a) Interchanging any two rows (or columns) this transformation is indicated as

$R_i \leftrightarrow R_j \left(C_i \leftrightarrow C_j\right)$

b) Multiplication of the elements of any row (or columns) by a non-zero scalar quantity this could be indicated as

$R_i \to kR_i \left(C_i \to kC_i\right)$

c) Addition of a constant multiple of the elements of any row to the corresponding element of any other row, indicated as $R_i \to R_i + kR_j$.

- **Equivalent Matrices :** Two matrices A and B are said to be equivalent, if one is obtained from the other by elementary transformations. It is denoted by $A \sim B$

- **Solution of Linear Equations by Determinants :**

1. **Cramer's Rule:** (Solution of system of linear equations in **two** unknowns) The solution of the system of equations

$a_1x + b_1y = c_1,\ a_2x + b_2y = c_2$

is given by $x = \dfrac{\Delta_1}{\Delta}$ and $y = \dfrac{\Delta_2}{\Delta}$, where

$$\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, \Delta_1 = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} and\ \Delta_2 = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

,provided $\Delta \neq 0$

2. **Cramer's Rule:** (Solution of system of

linear equations in **three** unknowns) The solution of the system of equations

$a_1x+b_1y+c_1z=d_1$, $a_2x+b_2y+c_2z=d_2$

$a_3x+b_3y+c_3z=d_3$ its matrix form is

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \Rightarrow AX=B$$

is given by $x=\frac{\Delta_1}{\Delta}, y=\frac{\Delta_2}{\Delta}$ *and* $z=\frac{\Delta_3}{\Delta}$, where

$$\Delta=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}; \Delta_1=\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix};$$

$$\Delta_2=\begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, \Delta_3=\begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

Provided $\Delta \neq 0$

Sub matrix: A matrix obtained by deleting finite number of rows or columns or both of a given matrix A is called sub matrix of A.

Rank of a matrix : Let A be a non - zero matrix. The rank of A is defined as the maximum of the orders of the non - singular square submatrices of A and is denoted by rank (A).

Note: i) If A is a non-zero matrix of order 3, then the rank of A is

a) 3 if A is non-singular

b) 2 if A is singular and there exist atleast one of its 2×2 submatrices is non-singular

c) 1 if A is singular and every 2×2 submatrix is singular

ii) A positive integer r is said to be the rank of a non-zero matrix A, if

a) There exists at least one minor in A of order r which is not zero. and

b) Every minor in A of order greater than r is zero. It is written rank (A) = r.

iii) The rank of a null matrix is defined as zero

iv) The rank of identy matrix of order n is n

v) If A is a non-singular matrix of order n then $rank(A)=n$.

vi) Elementary operations do not change the rank of a matrix.

vii) If A^T is a transpose of A, then

$\text{rank}(A^T)=\text{rank}(A)$

viii) If A^θ is the transposed conjugate of A, then

$rank(A^\theta)=rank(A)$.

ix) $rank(A+B)\le rank(A)+rank(B)$.

x) If A and B are two matrices such that the product AB is defined, then rank (AB) can not exceed the rank of the either matrix.

xi) If $A=\left[a_{ij}\right]_{m\times n}$ is a matrix of Rank r then

$r\le \min\{m,n\}$

- **Echelon form of a matrix :** In a matrix if the number of zeros before the first non zero element in any row doesnot exceed number of such type of zeros in the very next row, then the matrix is said to be in echelon form.

Eg : the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 1 & 3 & 5 & 1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ is in the Echelon form.

Note: The number of non zero rows of a matrix given in echelon form is its rank.

- **Characteristic Equation of a Matrix:**

If A is a matrix of order $n\times n$ then $|A-\lambda I|=0$ is called the characteristic equation of the matrix A.

If $A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, then its characteristic equation is $|A-\lambda I|=0$ takes the form

$\lambda^3-S_1\lambda^2+S_2\lambda-S_3=0$ where

$S_1=a_{11}+a_{22}+a_{33}$ = sum of the leading diagonal elements of A.

$$S_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

= sum of the minors of the leading diagonals elements of A

$S_3 = |A|$ =determinant of the matrix A.

W.E-8 : ***If A=*** $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix}$ ***then*** $A^3 - 3A^2 - I =$

Sol. Characteristic equation of A is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 1 & 0 \\ 1 & 2-\lambda & 1 \\ 2 & 1 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)[-\lambda(2-\lambda)-1] - 1[-\lambda - 2] = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - I = 0 \ (or) \ A^3 - 3A^2 - I = 0$$

- **Solution of system of non-homogenoeous linear equations in two unknowns :**

$a_1 x + b_1 y = c_1, \quad a_2 x + b_2 y = c_2$

Its matrix form is $\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$

$\Rightarrow AX = D$

i) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow$ system has unique solutions (consistent) $\Rightarrow X = A^{-1} D$

ii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow$ System has no solution (inconsistent)

iii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow$ System has infintely many solutions (consistent)

- **Solution of system of non-homogeneous linear equations in three unknowns :**

$a_1 x + b_1 y + c_1 z = d_1, a_2 x + b_2 y + c_2 z = d_2$

$a_3 x + b_3 y + c_3 z = d_3$ Its matrix form is

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \Rightarrow AX = D$$

(i) $|A| \neq 0$ then the system has unique solution

(ii) $|A| = 0$ then the system also has infinitely many solutions or no solution

Consistent System: A system of equations is said to be **consistent** if its solution (one or more) exist.

Inconsistent system: A system of equations is said to be **inconsistent** if its solution does not exist.

- **Solution of system of homogeneous linear equations in two unknowns :**

$a_1 x + b_1 y = 0, a_2 x + b_2 y = 0$

Its matrix form is $\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow AX = O$

(i) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow$ System has unique solution that is x = 0 = y (trivial solution)

(ii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \Rightarrow$ System has infinitely many solutions (non-trivial solutions)

(iii) The above system of equations is always consistent.

- **Solution of system of homogeneous linear equations in three unknowns :**

$a_1 x + b_1 y + c_1 z = 0, a_2 x + b_2 y + c_2 z = 0$

$a_3 x + b_3 y + c_3 z = 0$ Its matrix form is

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow AX = O$$

(i) $|A| \neq 0$ then the system has unique solution i.e $x = y = z = 0$ (trivial solution)

(ii) $|A| = 0$ then the system also has infinitely many solutions (non-trivial solutions)
(iii) The above system of equations is always consistent.

- **Solution of a system of Linear Equations by Matrix-Rank method :**

Let AX=B be a system of 'n' linear equations in 'n' variables.
1. Write the augmented matrix [A B]
2. Reduce the augmented matrix to echelon form using elementary row-opeations
3. Determine the rank of the coefficient matrix A and augmented matrix [A B] by counting the number of non-zero rows in A and [A B].

i) If $rank(A) \neq rank(AB)$,then the system of equations is inconsistent.

ii) If $rank(A) = rank(AB) =$ the number of unknowns, then the system of equations is consistent and has a unique solution.

iii) If $rank(A) = rank(AB) <$ the number of unknowns, then the system of equations is consistent and has infinitely many solutions.

- Let $AX = O$ be a homogeneous system of linear equations.

a) If $rank(A) =$ number of variables, then $AX = O$ have a trivial solution i.e. zero solution.

b) If $rank(A) <$ number of variables, then $AX = O$ have a non-trivial solution. It will have infinitely many solutions.

- **Conditions for consistency :**

The following cases may arise:

(i) If $\Delta \neq 0$, or Rank (A) = Rank [AB] =3 then the system is consistent and has a unique solution, which is given by Cramer's rule:

$$x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$$

(ii) If $\Delta = 0$ and at least one of the determinants $\Delta_1, \Delta_2, \Delta_3$ is non -zero,(or) $\Delta = 0$ and $(adjA)B \neq 0$ (or) rank (A) $\neq$ rank $[AB]$ the given system is inconsistent i.e., it has no solution.

(iii) If $\Delta = 0$ and $\Delta_1 = \Delta_2 = \Delta_3 = 0$,

(or) $\Delta = 0$ and $(adjA)B = 0$

(or) rank (A) = rank $[AB] < 3$ then the system is consistent and dependent, and has infinitely many solutions.

W.E-9: ***If the system of equations $x + y + z = 1$, $x + 2y + 4z = n$ and $x + 4y + 10z = n^2$ are consistent.then the values of 'n' are__***

Sol. $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{vmatrix} = \mathbf{0}$ $\therefore$ system has no solution or infinitely many solutions. Given system is consistent $\therefore$ It has infinitely many solutions

$$\therefore \Delta_3 = 0 \quad \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & n \\ 1 & 4 & n^2 \end{vmatrix} = 0$$

$\Rightarrow$ (1 - 2) (2 - n) (n - 1) = 0 $\Rightarrow$ n = 1, 2.

W.E-10: ***If α_1, α_2 are roots of $ax^2 + bx + c = 0$, β_1, β_2 are roots of $px^2 + qx + r = 0$ and equations $\alpha_1 y + \alpha_2 z = 0, \beta_1 y + \beta_2 z = 0$ have a non trivial solution, then $\frac{b^2}{q^2} =$ _____***

Sol. $\alpha_1 y + \alpha_2 z = 0$ *and* $\beta_1 y + \beta_2 z = 0$ have a non trivial solution

$$\therefore \begin{vmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{vmatrix} = 0 \Rightarrow \alpha_1\beta_2 - \alpha_2\beta_1 = 0$$

$$\Rightarrow \frac{\alpha_1}{\alpha_2} = \frac{\beta_1}{\beta_2} \quad \Rightarrow \frac{\alpha_1 + \alpha_2}{\alpha_1 - \alpha_2} = \frac{\beta_1 + \beta_2}{\beta_1 - \beta_2}$$

$$\Rightarrow \frac{(\alpha_1 + \alpha_2)^2}{(\alpha_1 + \alpha_2)^2 - 4\alpha_1\alpha_2} = \frac{(\beta_1 + \beta_2)^2}{(\beta_1 + \beta_2)^2 - 4\beta_1\beta_2}$$

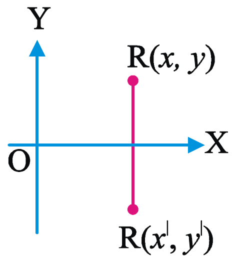
$$\Rightarrow \frac{\frac{b^2}{a^2}}{\frac{b^2}{a^2}-4\frac{c}{a}}=\frac{\frac{q^2}{p^2}}{\frac{q^2}{p^2}-\frac{4r}{p}}$$

$$\Rightarrow \frac{b^2}{b^2-4ac}=\frac{q^2}{q^2-4pr} \Rightarrow \frac{b^2}{q^2}=\frac{b^2-4ac}{q^2-4pr}$$

➢ Geometrical Transformations :

Reflection of a point $R(x,y)$ in x-axis :

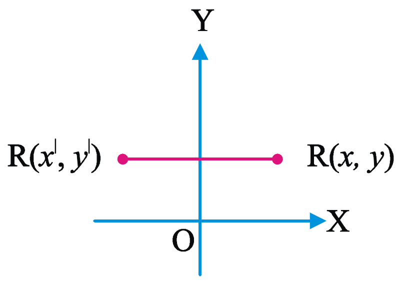
Let $R'(x',y')$ is the reflection of the point R (x,y) on x-axis. The matrix of reflection is given by



$$\begin{pmatrix} x' \\ y' \end{pmatrix}=\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

Reflection of a point $R(x,y)$ in y-axis:

$$R(x',y')=\begin{pmatrix} -1 & 0 \\ 0 & +1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$



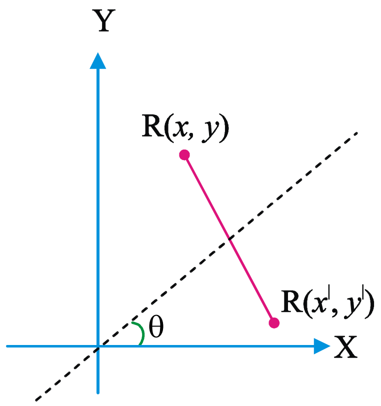
Note:(i) Reflection of a point $R(x,y)$ through origin is given by

$$\begin{bmatrix} x^1 \\ y^1 \end{bmatrix}=\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

(ii) Reflection of point $R(x,y)$ in the line y=x is given by $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(iii) Reflection of point $R(x,y)$ in the line y=-x is given by $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$

(iv) Reflection of point $R(x,y)$ in $y=(\tan\theta)x$ i.e $y=mx$ or $y=(\tan\theta)x$ is



given by

$$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix}=\begin{pmatrix} \frac{1-\tan^2\theta}{1+\tan^2\theta} & \frac{2\tan\theta}{1+\tan^2\theta} \\ \frac{-2\tan\theta}{1+\tan^2\theta} & \frac{1-\tan^2\theta}{1+\tan^2\theta} \end{pmatrix}$$

$$=\begin{pmatrix} \frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2} \\ \frac{-2m}{1+m^2} & \frac{1-m^2}{1+m^2} \end{pmatrix}$$

(v) Rotation of axes through an angle 'θ' is given by $R'(x',y')=\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$

Standard Results :

1. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3+b^3+c^3-3abc)$

2. $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2$

3. $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b)(b-c)(c-a)$

4. $\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & bc \\ 1 & b^2 & ca \\ 1 & c^2 & ab \end{vmatrix} =$

$(a-b)(b-c)(c-a)(a+b+c)$

5. $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = \begin{vmatrix} a & a^2 & bc \\ b & b^2 & ca \\ c & c^2 & ab \end{vmatrix}$

$= (a-b)(b-c)(c-a)(ab+bc+ca)$

6. $\begin{vmatrix} 1 & a & a^4 \\ 1 & b & b^4 \\ 1 & c & c^4 \end{vmatrix} = (a-b)(b-c)(c-a)$

$(a^2+b^2+c^2+ab+bc+ca)$

7. $\begin{vmatrix} n^2 & (n+1)^2 & (n+2)^2 \\ (n+1)^2 & (n+2)^2 & (n+3)^2 \\ (n+2)^2 & (n+3)^2 & (n+4)^2 \end{vmatrix} = -8$

8. $\begin{vmatrix} n^2 & (n+1)^2 & (n+2)^2 \\ (n+3)^2 & (n+4)^2 & (n+5)^2 \\ (n+6)^2 & (n+7)^2 & (n+8)^2 \end{vmatrix} = -216$

W.E-11: *If* $\begin{vmatrix} x+a & a^2 & a^3 \\ x+b & b^2 & b^3 \\ x+c & c^2 & c^3 \end{vmatrix} = 0$ ***and*** $a \neq b \neq c$

then $x =$

Sol: By given result

$$\begin{vmatrix} x & a^2 & a^3 \\ x & b^2 & b^3 \\ x & c^2 & c^3 \end{vmatrix} + \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} = 0$$

$$x\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} + abc\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = 0$$

$x(a-b)(b-c)(c-a)(ab+bc+ca)+$

$abc(a-b)(b-c)(c-a) = 0$

$$x = \frac{-abc}{(a-b)(b-c)(c-a)} \quad (\because a \neq b \neq c)$$

NCERT

Algebra Of Matrices

1. If two matrices A an B are of order $p \times q$ and $r \times s$ respectively, can be substracted only, if

1) $p = q$ 2) $p = q, r = s$

3) $p = r, q = s$ 4) $p = r$

2. A square matrix (a_{ij}) in which $a_{ij} = 0$ for $i \neq j$ and $a_{ij} = k$ (constant) for i = j is (EAM-2001)

1) Unit matrix 2) Scalar matrix

3) Null matrix 4) Diagonal matrix

3. If A = [a_{ij}] is a scalar matrix of order n × n, such that a_{ij} = k for all i=j, then trace of A =

1) nk 2) n+k 3) $\frac{n}{k}$ 4) 1

4. If A and B are two matrices such that A has identical rows and AB is defined. Then AB has

1) no identical rows 2) identical rows

3) no identical columns 4) cannot be determined

5. If AB = O, then

1) A = O 2) B = O

3) A and B need not be zero matrices

4) A and B are zero matrices

6. If A , B are two square matrices of order n and A and B commute (K be a real number). Then

1) $A - KI$, $B - KI$ Commute
2) $A - KI$, $B - KI$ are equal
3) $A - KI$, $B - KI$ do not commute
4) $A + KI$, $B - KI$ do not commute

7. If D_1 and D_2 are two 3×3 diagonal matrices then
1) $D_1 D_2$ is a diagonal matrix
2) $D_1 + D_2$ is a diagonal matrix
3) $D_1^2 + D_2^2$ is a diagonal matrix
4) 1, 2, 3 are correct

8. If AB = AC then
1) B = C   2) B $\neq$ C
3) B need not be equal to C   4) B = -C

9. If $AB = AC \Rightarrow B = C$, then A is
1) non-singular   2) singular
3) symmetric   4) Skew symmetric

10. If A and B are two matrices such that A + B and AB are both defined then
1) A and B are two matrices not necessarily of same order
2) A and B are square matrices of same order
3) A and B are matrices of same type
4) A and B are rectangular matrices of same order

11. If A = $\begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix}$ then A is
1) Nilpotent   2) involutary
3) Symmetric   4) Idempotent

12. A skew - symmetric matrix S satisfies the relation $S^2 + I = 0$, where I is a unit matrix then S is
1) Idempotent   2) Orthogonal
3) involutary   4) Nilpotent

13. If $A = \begin{bmatrix} 1 & 2-3i & 3+4i \\ 2+3i & 0 & 4-5i \\ 3-4i & 4+5i & 2 \end{bmatrix}$ then A is
1) Hermitian   2) Skew-Hermitian
3) Symmetric   4) Skew-Symmetric

14. If A and B are square matrices of size $n \times n$ such that $A^2 - B^2 = (A-B)(A+B)$, then which of the following will be always true.
1) AB=BA
2) Either A or B is a zero matrix
3) Either A or B is an identity matrix
4) A=B

15. If B is an idempotent matrix and $A = I - B$ then $AB =$
1) I 2) 0   3) $-I$ 4) B

16. If A is a symmetric or skew-symmetric matrix then A^2 is
1) symmetric   2) skew-symmetric
3) Diagonal   4) scalar or matrix

17. Let A be a square matrix. consider
1) $A + A^T$ 2) AA^T 3) A^TA 4) $A^T + A$
5) $A - A^T$ 6) $A^T - A$, Then
1) all are symmetric matrices
2) (2),(4),(6) are symmetric matrices
3) (1),(2),(3),(4) are symmetric matrices & (5),(6) are skew symmetric matrices
4) 5,6 are symmetric

18. If A, B are symmetric matrices of the same order then AB-BA is [EAM-2009]
1) symmetric matrix
2) skew symmetric matrix
3) Diagonal matrix   4) identity matrix

19. If a matrix A is both symmetric and skew-symmetric then A is
1) I   2) O
3) Both 1 and 2   4) Diagonal matrix

20. If A is a skew-symmetric matrix and n is odd positive integer, then A^n is
1) a symmetric matrix
2) skew-symmetric matrix
3) diagonal matrix   4) triangular matrix

21. If A is a skew-symmetric matrix and n is even positive integer , then A^n is
1) a symmetric matrix
2) skew-symmetric matrix
3) diagonal matrix   4) triangular matrix

22. If $A = \left[a_{ij}\right]_{3\times3}$ is a square matrix so that $a_{ij} = i^2 - j^2$, then A is a
1) unit matrix   2) symmetric marix
3) skew symmetric matrix   4) orthogonal matrix

23. If A, B are two idempotent matrices and

AB = B A = 0 then A+B is

1) Scalar matrix 2) Idempotent matrix
3) Diagonal matrix 4) Nilpotent matrix

24. **If A is a skew-symmetric matrix of order n, and C is a column matrix of order $n \times 1$ then $C^T AC$ is**
1) A Identity matrix of order n
2) A unit matrix of order one
3) A zero matrix of order one
4) A zero matrix of order n.

Determinants and Inverse of a Matrix :

25. **A and B are square matrices of order 3×3, A is an orthogonal matrix and B is a skew symmetric matrix. Which of the following statement is not true**

1) $|A| = \pm 1$ 2) $|B| = 0$ 3) $|AB| = 1$ 4) $|AB| = 0$

26. If $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$ **and A_i, B_i, C_i, are cofactors of a_i, b_i, c_i then $a_1 B_1 + a_2 B_2 + a_3 B_3 =$**

1) 0 2) $|A|$ 3) $|A|^2$ 4) $2|A|$

27. **If each element of a row of square matrix is doubled, the determinant of the matrix is**
1) not changed 2) doubled
3) multiplied by4 4) multiply by 1/2

28. **If A is a 3×3 singular matrix then A(Adj A) =**

1) $|A|$ I 2) I 3) O 4) ± 1

29. **If A and B are two non-singular matrices then**

$|\text{Adj}(AB)| =$

1) $|\text{Adj}(B)||\text{Adj}A|$ 2) $|\text{Adj}A||\text{Adj}B|$

3) Both (1) and (2) 4) $|A||B|$

30. **A_{3x3} is a nonsingular matrix $\Rightarrow A^2(\text{Adj}A) =$**

1) $|A|A$ 2) I 3) $|A|I$ 4) $|A|^2 I$

31. **If 'd' is the determinant of a square matrix A of order n, then the determinant of its adjoint is (EAM -2000)**
1) d^n 2) d^{n-1} 3) d^{n-2} 4) d

32. **If A is a non singular matrix then which of the following is not true**

1) $AdjA = |A|A^{-1}$ 2) $(AdjA)^{-1} = \frac{1}{|A|}A$

3) $\det(A^{-1}) = (\det A)^{-1}$ 4) $AdjA = I$

33. **If** $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ **then** $\begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} =$

$(A_1, A_2, A_3 \ldots \text{are cofactors})$

1) $\frac{\Delta^2}{2}$ 2) 2Δ 3) Δ^2 4) Δ

34. **Given $a_i^2 + b_i^2 + c_i^2 = 1 (i = 1,2,3)$ and**

$a_i a_j + b_i b_j + c_i c_j = 0 \ (i \neq j, i, j = 1,2,3)$ **then**

the value of $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ **is**

1) 0 2) $\frac{1}{2}$ 3) ± 1 4) 2

35. **If A, B are square matrices of order 3, then**

1) $AB = 0 \Rightarrow |A| = 0 \ and \ |B| = 0$

2) $AB = 0 \Rightarrow |A| = 0 \ or \ |B| = 0$

3) $(A+B)^{-1} = A^{-1} + B^{-1}$

4) adj (AB) = (adjA) (adjB)

36. **If the product of two non zero square matrices A and B of the same order is a zero matrix then**
1) Both are singular
2) atleast one of A&B is singular
3) A is non-singular, but B is singular
4) A is singlar but B is non-singular

37. **If A is an orthogonal matrix, then A^{-1} equals**

1) A 2) A^T 3) AA^T 4) I

38. **The inverse of a symmetric matrix is (if exists)**
1) Diagonal matrix 2) Symmetric matrix
3) Skew - symmetric matrix
4) Can't say

39. **The inverse of a skew symmetric matrix (if it exists) is**
1) a symmetric matrix
2) a skew symmetric matrix
3) a diagonal matrix 4) none of a matrix is unique

40. **The inverse of a skew symmetric matrix of odd order is**
1) a symmetric matrix
2) a skew symmetric matrix
3) diagonal matrix 4) does not exist

41. **If A is a square matrix of order 3 then**
$\left|Adj\ (Adj\ A^2)\right| =$
1) $|A|^2$ 2) $|A|^4$ 3) $|A|^8$ 4) $|A|^{16}$

Linear Equations :

42. **The system of equations which can be solved by cramer's rule have**
1) unique solution 2) no solution
3) infinitely many solutions 4) two solutions

43. **Let AX = B be a system of non homogeneous equations and det A = 0 then the system has**
1) infinity solutions 2) unique solution
3) no solution 4) infinity solutions or no solution

44. **The system AX = B of (n –a) equations in (n –a) unknowns has infinitely many solutions if**
1) det A $\neq$ 0, (adj A) (B) = 0
2) $|A| = 0, (adj\ A)(B) = 0$
3) $|A| = 0, (adj\ A)(B) \neq 0$ 4) Identity matrix

45. **The system of equations which can be solved by matrix inversion method have**
1) unique solution 2) no solution
3) infinitely many solutions 4) two solutions

46. **consider the system of equations**
$a_i x + b_i y + c_i z = 0$ **(where i=1,2,3), if**
$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$, **then the system has**
1) only one solution
2) one solution (0,0,0) and one more solution
3) no solution 4) infinite solutions

47. **If A is invertible matrix and B is another matrix such that (AB) exists then**
1) rank (AB) = rank (A)
2) rank (AB) = rank (B)
3) rank (AB) > rank (A)
4) rank (AB) > rank (B)

48. **If A is a non zero column matrix of order $m \times 1$ amd B is a non zero row matrix of order $1 \times n$ then rank of AB is**
1) 0 2)1 3) m 4) n

49. **If $A_{3\times3}.X_{3\times1} = D_{3\times1}$ is a consistent system of equations having unique solution then rank (A)**
1) 3 2) 2 3) 1 4) 0

50. **If A=$\left[a_{ij}\right]_{mxn}$ is a matrix of rank r then**
1) r = min {m, n} 2) r < min{m, n}
3) r $\leq$ min {m, n} 4) r = max {m, n}

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01) 3	02) 2	03) 1	04) 2	05) 3
06) 1	07)4	08) 3	09) 1	10) 2
11) 3	12) 2	13) 1	14) 1	15) 2
16) 1	17) 3	18) 2	19) 2	20) 2
21) 1	22) 3	23) 2	24) 3	25) 3
26) 1	27) 2	28) 1	29) 3	30) 1
31) 2	32) 4	33) 3	34) 3	35) 2
36) 1	37) 2	38) 2	39) 2	40) 4
41) 3	42) 1	43) 4	44) 2	45) 1
46) 4	47) 2	48) 2	49) 1	50) 3

LEVEL - I

Formation of Matrices:

1. **For 2×3 matrix $A = \left[a_{ij}\right]$ whose elements are given by $a_{ij} = \dfrac{(i+j)^2}{2}$ then A is equal to**

1) $\begin{bmatrix} 2 & 8 & \frac{9}{2} \\ 8 & \frac{9}{2} & \frac{25}{2} \end{bmatrix}$ 2) $\begin{bmatrix} 2 & \frac{9}{2} & 8 \\ \frac{9}{2} & 8 & \frac{25}{2} \end{bmatrix}$

3) $\begin{bmatrix} 2 & \frac{9}{2} & 8 \\ 8 & \frac{9}{2} & \frac{25}{2} \end{bmatrix}$ 4) $\begin{bmatrix} 2 & \frac{25}{2} & 8 \\ \frac{9}{2} & \frac{9}{2} & 8 \end{bmatrix}$

2. **If** $\begin{bmatrix} r+2 & 5 \\ -2 & r+1 \end{bmatrix} = \begin{bmatrix} 4 & y+3 \\ z & 3 \end{bmatrix}$**, then**

1) $r = y = z$ 2) $r = -y = z$
3) $-r = y = z$ 4) $r = y = -z$

Sum and Difference of the Matrices and Scalar Multiple of a Matrix:

3. **If A-2B =** $\begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}$ **and 2A-3B =** $\begin{pmatrix} -3 & 3 \\ 1 & -1 \end{pmatrix}$

then B =

1) $\begin{pmatrix} -5 & 7 \\ 5 & 1 \end{pmatrix}$ 2) $\begin{pmatrix} -5 & 7 \\ -5 & -1 \end{pmatrix}$

3) $\begin{pmatrix} -5 & 7 \\ 5 & -1 \end{pmatrix}$ 4) $\begin{pmatrix} -5 & -7 \\ -5 & -1 \end{pmatrix}$

4. **If I =** $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$**, B =** $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ **and**

$C = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$ then C=

1) $I\cos\theta + B\sin\theta$ 2) $I\sin\theta + B\cos\theta$
3) $I\cos\theta - B\sin\theta$ 4) $-I\cos\theta + B\sin\theta$

5. **If** $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}, B = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}$ **and 2A+C=B**

then C =

1) $\begin{bmatrix} 6 & 7 \\ 4 & 12 \end{bmatrix}$ 2) $\begin{bmatrix} 2 & -1 \\ -4 & -8 \end{bmatrix}$

3) $\begin{bmatrix} 3 & -1 \\ -4 & 8 \end{bmatrix}$ 4) $\begin{bmatrix} -6 & -7 \\ -4 & -12 \end{bmatrix}$

6. **If A=**$\begin{pmatrix} 0 & 2 \\ 3 & -4 \end{pmatrix}$**,kA=**$\begin{pmatrix} 0 & 3a \\ 2b & 24 \end{pmatrix}$ **then the values of** k, a, b **are respectively.**

1) -6, -12, -18 **2) -6, 4, 9**
3) -6, -4, -9 **4) -6, 12, 18**

Multiplication of Matrices:

7. **If (1 2 3) B = (3 4) then order of the matrix B is**

1) 3×1 **2)** 3×2 **3)** 2×4 **4)** 5×2

8. **If a matrix has 13 elements, then the possible dimensions (orders) of the matrix are**

1) 1 × 13 or 13 × 1 2) 1 × 26 or 26 × 1
3) 2 × 13 or 13 × 2 4) 13 × 13

9. **If** $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}$ **and** $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$ **then**

1) AB, BA exist and equal
2) AB, BA exist and are not equal
3) AB exist and BA does not exist
4) AB does not exist and BA exist

10. **If** $\begin{bmatrix} m & n \end{bmatrix}\begin{bmatrix} m \\ n \end{bmatrix} = \begin{bmatrix} 25 \end{bmatrix}$ **and** $m < n$**, then**

$(m,n) =$

1) $(2,3)$ 2) $(3,4)$ 3) $(4,3)$ 4) $(3,2)$

11. **If** $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$ **and** $A^n = 0$**, then the minimum value of 'n' is**

1) 2 2) 3 3) 4 4) 5

12. **If** $A = [x, y], B = \begin{bmatrix} a & h \\ h & b \end{bmatrix}, C = \begin{bmatrix} x \\ y \end{bmatrix}$**, then ABC=**

1) $(ax + hy + bxy)$ 2) $(ax^2 + 2hxy + by^2)$

3) $(ax^2 - 2hxy + by^2)$ 4) $(bx^2 - 2hxy + ay^2)$

13. **If A = diagonal (3,3,3) then** $A^4 =$

1)12A 2) 81A 3) 684A 4) 27A

14. **If** A=$\begin{bmatrix} 1 & -2 \\ 4 & 5 \end{bmatrix}$ **and** $f(t) = t^2 - 3t + 7$ **then**

f(A)+$\begin{bmatrix} 3 & 6 \\ -12 & -9 \end{bmatrix}$**=** **(EAM-2008)**

1) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ 2) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ 3) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ 4) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$

15. **If A =** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ **then** $\mathbf{A}^5 =$

1) I 2) O 3) A 4) A^2

16. If $A=\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix}$,then A^2-4A is equal to

1) $2I_3$ 2) $3I_3$ 3) $4I_3$ 4) $5I_3$

17. If $\begin{pmatrix}1&-\tan\theta\\\tan\theta&1\end{pmatrix}\begin{pmatrix}1&\tan\theta\\-\tan\theta&1\end{pmatrix}^{}=\begin{pmatrix}a&-b\\-b&a\end{pmatrix}$

1) $a=1,b=-1$ 2) $a=\sec^2\theta,b=0$

3) $a=0,b=\sin^2\theta$ 4) $a=\sin 2\theta,b=\cos 2\theta$

18. If $A=\begin{bmatrix}a^2&ab&ac\\ab&b^2&bc\\ac&bc&c^2\end{bmatrix}$ and $a^2+b^2+c^2=1$,

then $A^2=$

1) A 2) 2A 3) 3A 4) 4A

19. A fruit shop has 5 dozen oranges, 3 dozen mangoes, 6 dozen bananas their selling prices are Rs 60, Rs40, Rs30 each respectively. Using matrix algebra the value of the fruits in the shop is

1) 7200 2) 7000 3) 2700 4) 7500

Problems based on Induction:

20. If $A=\begin{bmatrix}x&x\\x&x\end{bmatrix}$, then $A^n=$ ...., $n\in N$

1) $\begin{bmatrix}2^nx^n&2^nx^n\\2^nx^n&2^nx^n\end{bmatrix}$ 2) $\begin{bmatrix}2^{n-1}x^n&2^{n-1}x^n\\2^{n-1}x^n&2^{n-1}x^n\end{bmatrix}$

3) $\begin{bmatrix}2^{n-2}x^n&2^{n-2}x^n\\2^{n-2}x^n&2^{n-2}x^n\end{bmatrix}$ 4) $\begin{bmatrix}2^{n-1}x^{n-1}&2^{n-1}x^{n-1}\\2^{n-1}x^{n-1}&2^{n-1}x^{n-1}\end{bmatrix}$

21. Matrix A is such that $A^2 = 2A - I$ where I is the unit matrix . Then for $n \geq 2$, $A^n =$ (EAM-2013)

1) $nA-(n-1)I$ 2) $nA-I$

3) $2^{n+1}A-(n-1)I$ 4) $2^{n+1}A-1$

22. $\begin{bmatrix}i&o&o\\o&i&o\\o&o&i\end{bmatrix}$ then $A^{4n+1}=\ldots, n\in N$

1) $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$ 2) $\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&-1\end{bmatrix}$

3) $\begin{bmatrix}i&0&0\\0&i&0\\0&0&i\end{bmatrix}$ 4) $\begin{bmatrix}-i&0&0\\0&-i&0\\0&0&-i\end{bmatrix}$

23. If $A=\begin{bmatrix}1&0\\1&1\end{bmatrix}$ and $I=\begin{bmatrix}1&0\\0&1\end{bmatrix}$, then which one of the following holds for all $n\geq 1$ (by the principle of mathematical induction)

1) $A^n=nA+(n-1)I$

2) $A^n=2^{n-1}A+(n-1)I$

3) $A^n=nA-(n-1)I$

4) $A^n=2^{n-1}A-(n-1)I$

Trace of Matrix:

24. If Tr(A) = 6 $\Rightarrow$ Tr (4A)=

1) 3/2 2) 2 3) 12 4) 24

25. If Tr (A)=3, Tr(B)=5 then Tr(AB) =

1)15 2)5 3)3/5 4) Cannot say

26. If $A=\begin{bmatrix}1&2&3\\4&5&6\\7&1&0\end{bmatrix}$, $B=\begin{bmatrix}1&0&0\\0&3&0\\0&4&5\end{bmatrix}$, $Tr(BA)=\ldots$

1) 40 2) 45 3) 39 4) 5

Transpose and Properties of Transpose of Matrix:

27. If A is a 3 × 4 matrix and B is matrix such that A^TB and BA^T are both define then order of B is

1) 3 × 4 2) 4 × 3 3) 3 × 3 4) 4 × 4

28. $\begin{bmatrix}r+4&6\\3&r+3\end{bmatrix}=\begin{bmatrix}5&r+2\\r+5&4\end{bmatrix}^T$ then $r=$

1) 1 2) 2 3) 3 4) -1

29. If 3A + 4B^T= $\begin{pmatrix}7&-10&17\\0&6&31\end{pmatrix}$ and

$2B-3A^T = \begin{pmatrix} -1 & 18 \\ 4 & -6 \\ -5 & -7 \end{pmatrix}$ then B =

1) $\begin{pmatrix} 1 & 3 \\ -1 & 0 \\ -2 & -4 \end{pmatrix}$ 2) $\begin{pmatrix} 1 & 3 \\ 1 & 0 \\ 2 & 4 \end{pmatrix}$

3) $\begin{pmatrix} 1 & 3 \\ -1 & 0 \\ 2 & 4 \end{pmatrix}$ 4) $\begin{pmatrix} 1 & -3 \\ 1 & 0 \\ 2 & 4 \end{pmatrix}$

30. If $5A = \begin{pmatrix} 3 & -4 \\ 4 & x \end{pmatrix}$ and $AA^T = A^T A = I$ then x=

1) 3 2) -3 3) 2 4) -2

Symmetric, Skew Symmetric Matrices & Special types of Matrices:

31. If $A = \begin{bmatrix} 2 & x-3 & x-2 \\ 3 & -2 & -1 \\ 4 & -1 & -5 \end{bmatrix}$ is a symmetric matrix then $x =$

1) 0 2) 3 3) 6 4) 8

32. $\begin{pmatrix} 2 & 3 & 5 \\ 4 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix} = P + Q$, where P is a symmetric and Q is a skew-symmetric then Q=

1) $\begin{pmatrix} 0 & \frac{-1}{2} & 2 \\ \frac{1}{2} & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix}$ 2) $\begin{pmatrix} 0 & \frac{1}{2} & 1 \\ \frac{-1}{2} & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}$

3) $\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$ 4) $\begin{pmatrix} 0 & 2 & 3 \\ -2 & 0 & 4 \\ -3 & -4 & 0 \end{pmatrix}$

33. If $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ then A is

1) an idempotent marix 2) nilpotent matrix
3) involutary 4) orthogonal matrix

Determinants and its Properties:

34. If $A = \begin{bmatrix} a & c & b \\ b & a & c \\ c & b & a \end{bmatrix}$ then the cofactor of a_{32} in $A + A^T$ is

1) $-\left(2a(b+c)-(b+c)^2\right)$ 2) $ac-b^2$

3) a^2-bc 4) $2a(a+c)-(a+c)^2$

35. The value of $\begin{vmatrix} 2+i & 2-i \\ 1+i & 1-i \end{vmatrix}$ is $\{\because i^2 = -1\}$

1) A complex quantity 2) real quantity
3) 0 4) cannot be determined

36. det $\begin{bmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{bmatrix} = ...$

1) -8 2) -6 3) -1 4) 0

37. $\begin{vmatrix} 24 & 25 & 26 \\ 25 & 26 & 27 \\ 26 & 27 & 27 \end{vmatrix}$ is equal to **(EAM-2011)**

1) 0 2) -1 3) 1 4) 2

38. $\begin{vmatrix} b^2+c^2 & a^2 & a^2 \\ b^2 & c^2+a^2 & b^2 \\ c^2 & c^2 & a^2+b^2 \end{vmatrix} =$

1) $a^2b^2c^2$ 2) 4abc 3) $4a^2b^2c^2$ 4) $2a^2b^2c^2$

39. $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} =$

(EAM-2008,1995)

1) $2(a+b+c)^3$ 2) $(a-b-c)^3$
3) $2(a-b-c)^3$ 4) $(a+b+c)^3$

40. If $\begin{vmatrix} \lambda^2+3\lambda & \lambda-1 & \lambda+3 \\ \lambda+1 & 2-\lambda & \lambda-4 \\ \lambda-3 & \lambda+4 & 3\lambda \end{vmatrix}$

$= p\lambda^4+q\lambda^3+r\lambda^2+s\lambda+t$ then $t=$

1) 16 2) 17 3) 18 4) 19

41. $A=\begin{bmatrix} 5 & 5\alpha & \alpha \\ 0 & \alpha & 5\alpha \\ 0 & 0 & 5 \end{bmatrix}$;If $|A^2|=25$ .then $|\alpha|=$

[EAM-2007]

1) 5 2) 5^2 3) 1 4) $\frac{1}{5}$

42. If a, b, c are in A.P.then

$\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix} =$

1) 1 2) 0 3) -1 4) 2

43. If x,y, z are all different and if

$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$ then $1+xyz=$

(EAM-2010)

1) -1 2) 0 3) 1 4) 2

44. $\begin{vmatrix} x+2 & x+3 & x+5 \\ x+4 & x+6 & x+9 \\ x+8 & x+11 & x+15 \end{vmatrix} =$ **(EAM-2013)**

1) $3x^2+4x+5$ 2) x^2+8x+2
3) 0 4) -2

45. If $a\neq 6, b, c$ satisfy $\begin{vmatrix} a & 2b & 2c \\ 3 & b & c \\ 4 & a & b \end{vmatrix} = 0$, then $abc=$

1) $a+b+c$ 2) 0
3) b^3 4) $ab+b-c$

46. $\begin{vmatrix} \log e & \log e^2 & \log e^3 \\ \log e^2 & \log e^3 & \log e^4 \\ \log e^3 & \log e^4 & \log e^5 \end{vmatrix} =$

1) 0 2) 1 3) 4 loge 4) 5 loge

47. If $\begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = 0$ then $x=$

1) a 2) b 3) a or b 4) 0

48. **The value of a third order determinant is 11, then the value of the square of the determinant formed by the cofactors will be**

1) 11 2) 121 3) 1331 4) 14641

49. **If matrix A is an circulant matrix whose elements of first row are a, b, c all >0 such that abc = 1 and** $A^TA=I$ **then** $a^3+b^3+c^3$ **equals**

1) 0 2) 4 3) 1 4) 3

50. **In a third order determinant, each element of the first column consists of sum of two terms, each element of the second column consists of sum of three terms and each element of the third column consits of sum of four terms,.Then it can be decomposed into n determinants, where n has value**

1) 1 2) 9 3) 16 4) 24

Adjoint of a matrix:

51. If A $=\begin{bmatrix} 1^2 & 2^2 & 3^2 \\ 2^2 & 3^2 & 4^2 \\ 3^2 & 4^2 & 5^2 \end{bmatrix}$ then $|Adj\ A| =$

1) 8 2) 16 3) 64 4) 128

52. **If A is a square matrix such that A (AdjA)**

$=\begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$ **then det (AdjA) = (EAM-2007)**

1) 4 2) 16 3) 64 4) 256

53. Adj $\begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & -2 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & a & -2 \\ 1 & 1 & 0 \\ -2 & -2 & b \end{bmatrix} \Rightarrow [a\ \ b] =$

1) $[-4\ \ 1]$ 2) $[-4\ \ -1]$

3) $\begin{bmatrix} 4 & 1 \end{bmatrix}$ 4) $\begin{bmatrix} 4 & -1 \end{bmatrix}$

54. **If A is a 3x3 matrix and $|adjA| = 16$ then $|A| =$**

1) 4 2) -4 3) ± 4 4) 8

55. **If det $(A_{3\times3}) = 6$, then det (adj 2A)=**

1) 144 2) $2^2 \times 3^8$ 3) $3^3 \times 2^4$ 4) $3^2 \times 2^8$

Inverse of a Matrix:

56. **If $\begin{bmatrix} 1 & -1 & x \\ 1 & x & 1 \\ x & -1 & 1 \end{bmatrix}$ has no inverse, then the real value of x is** **[EAM-2009]**

1) 2 2) 3 3) 0 4) 1

57. **If $\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix} \begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix}^{-1} =$**

$\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}^{-1}$ **then $\alpha =$**

1) 0 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{6}$

58. **The inverse of**

$\begin{bmatrix} 1 & a & b \\ 0 & x & 0 \\ 0 & 0 & 1 \end{bmatrix} is \begin{bmatrix} 1 & -a & -b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} then\ x =$

1) a 2) b 3) 0 4) 1

59. **If a, b, c and d are real numbers such that and if $A = \begin{bmatrix} a+ib & c+id \\ -c+id & a-ib \end{bmatrix}$ then $A^{-1} =$** **[EAM-2014]**

1) $\begin{bmatrix} a+ib & -c-id \\ c-id & a-ib \end{bmatrix}$ 2) $\begin{bmatrix} a-ib & c+id \\ -c+id & a+ib \end{bmatrix}$

3) $\begin{bmatrix} a-ib & -c-id \\ c-id & a+ib \end{bmatrix}$ 4) $\begin{bmatrix} a+ib & c+id \\ c-id & a-ib \end{bmatrix}$

60. **If A is a non zero square matrix of order n with det(I+A) $\neq$ 0 and $A^3 = 0$, where I, O are unit and null matrices of order nxn respectively then $(I + A)^{-1} =$** **(EAM-2010)**

1) $I - A + A^2$ 2) $I + A + A^2$

3) $I + A^2$ 4) $I + A$

61. **If $|A| \neq 0$ and $(A-2I)(A-3I) = 0$ then $A^{-1} =$**

1) $\frac{A-5I}{6}$ 2) $\frac{5I-A}{5}$ 3) $\frac{5A-I}{6}$ 4) $\frac{5I-A}{6}$

Rank of a Matrix:

62. **The rank of the matrix A = $\begin{bmatrix} -1 & 2 & 3 \\ -2 & 4 & 6 \end{bmatrix}$ is**

1) 3 2) 2 3) 1 4) 0

63. **If I is a (9×9) unit matrix, then rank (I) =**

1) 0 2) 3 3) 6 4) 9

64. **The ranks of the matrices in descending order**

A. $\begin{bmatrix} 1 & 4 & -1 \\ 2 & 3 & 0 \\ 0 & 1 & 2 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ **C.** $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{bmatrix}$

1) A,B,C 2) A,C,B 3) B,C,A 4) C,A,B

65. **If the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & \alpha \end{bmatrix}$ is of rank 3, then $\alpha =$** **(EAM-2014)**

1) -5 2) 5 3) 4 4) -4

Solution of Simultaneous Equation non Homogeneous Linear Equations:

66 **The solution of $2x + y + z = 1$, $x - 2y - 3z = 1$, $3x + 2y + 4z = 5$ is**

1) 1,2,3 2) 1,2,-3 3) 1,-3,2 4) 1,3,2

67. **The system of equations**

$2x + 6y + 11 = 0$, $6y - 18z + 1 = 0$

$6x + 20y - 6z + 3 = 0$

1) is consistent 2) has unique solution

3) is inconsistent 4) cannot be determined

68. **The value of 'a' for which the equations $3x - y + az = 1$, $2x + y + z = 2$, $x + 2y - az = -1$ fail to have unique solution is**

1) 7/2 2) -7/2 3) 2/7 4) -2/7

69. **The system of equations $3x + 2y + z = 6$, $3x + 4y + 3z = 14$ and $6x + 10y + 8z = a$, has**

infinite number of solutions, if a is equal to (EAM-2013)

1) 8 2) 12 3) 24 4) 36

70. The number of solutions of the equation
$3x+3y-z=5, x+y+z=3, 2x+2y-z=3$

1) 1 2) 0 3) infinite 4) two

Homogeneous Linear Equations:

71. The number of non-trivial solutions of the system $x-y+z=0, x+2y-z=0,$
$2x+y+3z=0$ **is [EAM-2007]**

1) 0 2) 1 3) 2 4) 3

72. If x, y, z not all zeros and the equations
$x=cy+bz, y=az+cx, z=bx+ay$
are consistent then a relation among a,b,c is (AIE-2008)

1) $a^2+b^2+c^2=0$ 2) $a^2+b^2+c^2-2abc=0$
3) $a^2+b^2+c^2+2abc=1$
4) $a^2+b^2+c^2+2abc=0$

73. If x, y, z not all zeros and the equations $x+y+z=0$, $(1+a)x+(2+a)y-8z=0$, $x-(1+a)y+(2+a)z=0$ **have non-trivial solution then a =**

1) $2+\sqrt{15}$ 2) $3\pm\sqrt{15}$
3) $\sqrt{15}$ 4) $-5\pm2\sqrt{2}$

74. If $a\neq b\neq c\neq 1$ and the system $ax+y+z=0$ $x+by+z=0$, $x+y+cz=0$ **have non trivial solutions then** $a+b+c-abc=$ ....

1) 0 2) 1 3) 2 4) 4

LEVEL - I - KEY

01) 2	02) 4	03) 2	04) 1	05) 2
06) 3	07) 2	08) 1	09) 2	10) 2
11) 1	12) 2	13) 4	14) 2	15) 3
16) 4	17) 2	18) 1	19) 1	20) 2
21) 1	22) 3	23) 3	24) 4	25) 4
26) 1	27) 1	28) 1	29) 3	30) 1
31) 3	32) 1	33) 1	34) 1	35) 1
36) 3	37) 3	38) 3	39) 4	40) 3
41) 4	42) 2	43) 2	44) 4	45) 3
46) 1	47) 3	48) 4	49) 2	50) 4
51) 3	52) 2	53) 3	54) 3	55) 4
56) 4	57) 1	58) 4	59) 3	60) 1
61) 4	62) 3	63) 4	64) 2	65) 2
66) 3	7) 3	68) 2	69) 4	70) 3
71) 1	72) 3	73) 4	74) 3	

LEVEL - I (C.W)-HINTS

1. $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$ be a 2×3 matrix find all elements.

2. Equating corresponding elements

3. $2A-4B=\begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$

$2A-3B=\begin{bmatrix} -3 & 3 \\ 1 & -1 \end{bmatrix}$ subtract and find B.

4. $C=\begin{pmatrix} \cos\theta & 0 \\ 0 & \cos\theta \end{pmatrix}+\begin{pmatrix} 0 & \sin\theta \\ -\sin\theta & 1 \end{pmatrix}$

$=I\cos\theta+B\sin\theta$

5. $C=B-2A$, $C=\begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix}-\begin{bmatrix} 2 & 4 \\ 4 & 10 \end{bmatrix}$

6. $A=\begin{bmatrix} \frac{0}{k} & \frac{3a}{k} \\ \frac{2b}{k} & \frac{24}{k} \end{bmatrix}$

7. order of B is 3×2

8. possible order of the matrices are 1×13 or 13×1

9. AB, BA exists and not equal

10. $m^2+n^2=25$ and m<n Hence $(m,n)=(3,4)$

11. $A^2=\begin{bmatrix} a^2b^2-a^2b^2 & ab^3-ab^3 \\ -a^3b+a^3b & -a^2b^2+a^2b^2 \end{bmatrix}=0$

$\Rightarrow A^3=A.A^2=0$ and $A^n=0$, for all $n\geq2$

12. find ABC

13. $A^4 = 81I = 27A$

14. $f(A) = A^2 - 3A + 7I$

15. $A^2 = I \therefore A^5 = A$

16. $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}, A^2 = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$

$\therefore A^2 - 4A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = 5I_3$

17. Use multiplication of matrices and equal corresponding elements.

18. Find A^2 and use $a^2 + b^2 + c^2 = 1$

19. No of fruits

$A = [5 \times 12 = 60 \;\; 3 \times 12 = 36 \;\; 6 \times 12 = 72]$

selling prices (in Rs)

$B = \begin{bmatrix} 60 \\ 40 \\ 30 \end{bmatrix}$ total value of the fruits

$AB = \begin{bmatrix} 60 & 36 & 72 \end{bmatrix} \begin{bmatrix} 60 \\ 40 \\ 30 \end{bmatrix} = [7200]$

20. put n = 3

21. put n = 3

22. put n = 1

23. $A^2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, A^3 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \therefore A^n = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix}$

$nA = \begin{bmatrix} n & 0 \\ n & n \end{bmatrix}, (n-1)I = \begin{bmatrix} n-1 & 0 \\ 0 & n-1 \end{bmatrix}$

$nA - (n-1)I = \begin{bmatrix} 1 & 0 \\ n & 1 \end{bmatrix} = A^n$

24.. $Tr(4A) = 4Tr(A)$

25. $Tr(AB) \neq Tr(A).Tr(B)$

26. Find BA

27. Verification

28. $r + 4 = 5 \Rightarrow r = 1$

29. find $\left(3A + 4B^T\right)^T$ and adding

30. find AA^T

31. $A^T = A$

32. $Q = \dfrac{A - A^T}{2}$

33. $A^2 = A$

34. find $A + A^T$ and find cofactor

35. expand

36. Applying $C_2 \to C_2 - 2C_1$

$C_3 \to C_3 - 2C_2$

37. $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_2$ we get $\begin{vmatrix} 24 & 25 & 26 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 1$

38. put a = 1, b = 1, c = 2 verify option

39. Put a=1,b=1,c=1

40. put $\lambda = 0$

41. $\Rightarrow |A| = 25\alpha \Rightarrow |A|^2 = 625\alpha^2$

42. Put a=1, b=2, c=3

43. Split into two determinants

44. $R_2 \to R_2 - R_1, R_3 \to R_3 - R_2$ and expand

45. expand

46. $\begin{vmatrix} \log e & 2\log e & 3\log e \\ 3\log e & 3\log e & 4\log e \\ 3\log e & 4\log e & 5\log e \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} (\log e)^3$

$R_2 \to R_2 - R_1, R_3 \to R_3 - R_2$

47. $m\begin{vmatrix} a & a & x \\ 1 & 1 & 1 \\ b & x & 6 \end{vmatrix} = 0$ and expand

we get $(b-x)(a-x)=0$

48. It is square of

$\det(adj\ A) = \left((\det A)^{n-1}\right)^2 = \left(11^2\right)^2 = 11^4$

$= 14641$

49. $A^T A = I \Rightarrow$ 'A' is a orthogonal matrix $\Rightarrow |A| = \pm 1$

$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = \pm 1 \Rightarrow 3abc - a^3 - b^3 - c^3 = \pm 1$

$\Rightarrow 3(1) \pm 1 = a^3 + b^3 + c^3$

$\Rightarrow a^3 + b^3 + c^3 = 4\ or\ 2$

50. 2 x 3 x 4 = 24

51. $|AdjA| = |A|^{n-1}$

52. $A(adjA) = |A|.I$

53. Find adjA

54. $|AdjA| = 16 \Rightarrow |A|^2 = 16 \Rightarrow |A| = \pm 4$

55. $|AdjkA| = |kA|^{n-1} = (k^n|A|)^{n-1}$

56. i.e $\begin{vmatrix} 1 & -1 & x \\ 1 & x & 1 \\ x & -1 & 1 \end{vmatrix} = 0$ and expand

57. $AA^{-1} = I$ and $I^{-1} = I$

58. vertify with $AA^{-1} = I$

59. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the $A^{-1} = \dfrac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

60. $(I^3 + A^3) = (I+A)(I - A + A^2) = I$

61. $A^2 - 5A + 6I = 0$

$5A = A^2 + 6I \because A^{-1} = \dfrac{5I - A}{6}$

62. All 2×2 sub matrices are singular

63. Rank of I is its order

64. Rank of the matrixof order 3×3

65. det A = 0

66. By verification

67. $\text{Rank}(A) \neq Rank(AD)$

68. $\text{Rank}(A) = Rank(AD) = 3$

69. $\text{Rank}(A) = Rank(AD) < 3$

70. $\text{Rank}(A) = Rank(AD) < 3$

71. $\begin{vmatrix} 1 & -1 & 1 \\ 1 & 2 & -1 \\ 2 & 1 & 3 \end{vmatrix} = 0$

72. $\begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0$

73. $\begin{vmatrix} 1 & 1 & 1 \\ 1+a & 2+a & -8 \\ 1 & -(1+a) & (2+a) \end{vmatrix} = 0$

74. $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$

LEVEL - II

Formation of Matrices:

1. If $a_{ij} = \frac{1}{2}(3i - 2j)$ and $A = [a_{ij}]_{2\times2}$, then A is equal to

1) $\begin{bmatrix} 1/2 & 2 \\ -1/2 & 1 \end{bmatrix}$ 2) $\begin{bmatrix} 1/2 & -1/2 \\ 2 & 1 \end{bmatrix}$

3) $\begin{bmatrix} 2 & 2 \\ 1/2 & -1/2 \end{bmatrix}$ 4) $\begin{bmatrix} -2 & -2 \\ -1/2 & 1/2 \end{bmatrix}$

2. If $\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4a-z \end{bmatrix} = \begin{bmatrix} 0 & -7 \\ 3 & 2a \end{bmatrix}$,

$(x+y+z+a)=$

1) -1 2) 0 3) 1 4) 8

Sum and Difference of the Matrices and scalar multiple of matrix:

3. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ **+ 2X =** $\begin{bmatrix} 3 & 5 \\ 5 & 9 \end{bmatrix}$, $\Rightarrow$ **X =**

1) $\begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix}$ 2) $\begin{bmatrix} 1 & 3/2 \\ 1 & 5/2 \end{bmatrix}$

3) $\begin{bmatrix} -2 & -3 \\ 2 & 5 \end{bmatrix}$ 4) $\begin{bmatrix} 1 & 3/2 \\ -1 & -5/2 \end{bmatrix}$

4. The additive inverse of $\begin{pmatrix} 1 & 4 & -7 \\ -3 & 2 & 5 \\ 2 & 3 & -1 \end{pmatrix}$ **is**

1) $\begin{pmatrix} -1 & -4 & 7 \\ 3 & 2 & -5 \\ 2 & 3 & -1 \end{pmatrix}$ 2) $\begin{pmatrix} -1 & -4 & 7 \\ 3 & -2 & -5 \\ -2 & -3 & -1 \end{pmatrix}$

3) not possible 4) $\begin{pmatrix} -1 & -4 & 7 \\ 3 & -2 & -5 \\ -2 & -3 & 1 \end{pmatrix}$

5. If $A=\begin{bmatrix} 9 & 1 \\ 4 & 3 \end{bmatrix}, B=\begin{bmatrix} 1 & 5 \\ 6 & 11 \end{bmatrix}$ **and**

3A+5B+2X=0 then X=

1) $\begin{bmatrix} 16 & -14 \\ 21 & -32 \end{bmatrix}$ 2) $\begin{bmatrix} 16 & 14 \\ -21 & -32 \end{bmatrix}$

3) $\begin{bmatrix} -16 & -14 \\ -21 & -32 \end{bmatrix}$ 4) $\begin{bmatrix} -16 & 14 \\ 21 & 32 \end{bmatrix}$

Multiplication of Matrices:

6. The order of [x y z] $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ **is**

1) 3×1 2) 1×1 3) 1×3 4) 3×3

7. If $A=\begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix}$ **then A^2 =**

1) $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ 2) $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$ 3) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ 4) $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

8. If $A=\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ **then** $A^5=$

1) 243 2) 81A 3) 243A 4) 81I

9. If $\begin{bmatrix} x & y & z \end{bmatrix}\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}=$

1) $\left[ax^2+by^2+cz^2+2hxy+2gxz+2fyz\right]$

2) $\left[ax^2+by^2+cz^2+hxy+gxz+fyz\right]$

3) $\left[2ax^2+2by^2+2cz^2+hxy+gxz+fyz\right]$

4) $\left[2ax^2+2by^2+2cz^2+2hxy+2gxz+2fyz\right]$

10. If AB = A and BA = B then

1) A = 2B 2) $A^2 = A$ and $B^2 = B$

3) 2A = B 4) cannot be determined

11. If $A=\begin{bmatrix} 2 & 1 \\ 3 & 0 \end{bmatrix}$ **then** $A^2+2A+I=$

1) $\begin{bmatrix} 12 & 4 \\ 12 & 4 \end{bmatrix}$ 2) $\begin{bmatrix} 12 & -4 \\ 4 & 12 \end{bmatrix}$

3) $\begin{bmatrix} 4 & 12 \\ 12 & 4 \end{bmatrix}$ 4) $\begin{bmatrix} 4 & 12 \\ -12 & -4 \end{bmatrix}$

12. If $I=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, E=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ **, then** $(aI+bE)^3=$

1) $aI+bE$ 2) a^3I+b^3E

3) a^3I+3ab^2E 4) a^3I+3a^2bE

13. If $A=\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$ then $A^{2016}=$

1) I 2) 0 3) A 4) A^2

14. If $P=\begin{bmatrix}1\\3\\4\end{bmatrix}$, $Q=\begin{bmatrix}2 & -1 & 5\end{bmatrix}$, then PQ =

1) $\begin{bmatrix}2 & -1 & 5\\6 & -3 & 15\\8 & -4 & 20\end{bmatrix}$ 2) $\begin{bmatrix}2 & -3 & 20\end{bmatrix}$

3) $\begin{bmatrix}2\\-3\\20\end{bmatrix}$ 4) $[19]$

15. If $A=\begin{bmatrix}o & c & -b\\-c & o & a\\b & -a & o\end{bmatrix}$ and $B=\begin{bmatrix}a^2 & ab & ac\\ab & b^2 & bc\\ac & bc & c^2\end{bmatrix}$

then AB =

1) A 2) B 3) I 4) O

Trace of Matrix:

16. If Tr (A) = 2+i $\Rightarrow$ Tr [(2-i)A] =

1) 2+i 2) 2-i 3) 3 4) 5

17 If Tr (A) = 8, Tr(B) = 6, $\Rightarrow$ Tr (A - 2B) =

1) -4 2) 4 3) 2 4) 11

Problems Based on Induction:

18. $\begin{pmatrix}2 & -1\\3 & -2\end{pmatrix}^n=\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$ if 'n' is

1) odd 2) any natural number
3) even 4) not possible

19. If $A=\begin{bmatrix}3 & -4\\1 & -1\end{bmatrix}$ then A^P where $P\in N$ is

1) $\begin{bmatrix}2P & -4P\\P & 1-2P\end{bmatrix}$ 2) $\begin{bmatrix}1+2P & -4P\\P & 1-2P\end{bmatrix}$

3) $\begin{bmatrix}1+2P & -4P\\P & -P\end{bmatrix}$ 4) $\begin{bmatrix}1+2p & -4\\-p & 1-2p\end{bmatrix}$

20. If $A=\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}$, $n\in N$ then $A^n=$

1) $2^{n-1}A$ 2) 2^nA 3) nA 4) 2n

Transpose and Properties of Transpose of Matrix:

21. If $O(A)=2\times3, O(B)=3\times2$, and $O(C)=3\times3$, which one of the following is not defined

1) $CB+A^T$ 2) BAC
3) $C(A+B^T)^T$ 4) $C(A+B^T)$

22. If the order of A is 4 × 3, the order of B is 4 × 5 and the order of C is 7 × 3, then the order of $(A^TB)^TC^T$ is

1) 4 × 5 2) 3 × 7 3) 4 × 3 4) 5 ×7

23. If $3A=\begin{bmatrix}-1 & 2 & 2\\2 & -1 & 2\\2 & 2 & -1\end{bmatrix}$ then

1) $AA^T=A^TA=I$ 2) $AA^T=A^TA=-I$
3) $AA^T=A^TA=0$ 4) $AA^T=A^TA=A$

24. Which of the following is not true, if A and B are two matrices each of order $n\times n$, then

1) $(A+B)^T=A^T+B^T$ 2) $(A-B)^T=A^T-B^T$
3) $(AB)^T=A^TB^T$ 4) $(ABC)^T=C^TB^TA^T$

Symmetric, Skew Symmetric & Special types of Matrices:

25. $A=\begin{bmatrix}x & -7\\7 & y\end{bmatrix}$ is a skew-symmetric matrix, then $(x,y)=$

1) (1,-1) 2) (7,-7) 3) (0,0) 4)(14,-14)

26. $\begin{bmatrix}1 & 6\\7 & 2\end{bmatrix}$=P+Q, where P is a symmetric & Q is a skew-symmetric then P=

1) $\begin{bmatrix}1 & \frac{13}{2}\\\frac{13}{2} & 2\end{bmatrix}$ 2) $\begin{bmatrix}1 & -\frac{13}{2}\\-\frac{13}{2} & 2\end{bmatrix}$

3) $\begin{bmatrix}1 & -\frac{1}{2}\\\frac{1}{2} & 2\end{bmatrix}$ 4) $\begin{bmatrix}0 & 13/2\\13/2 & 0\end{bmatrix}$

27. If $A=\begin{bmatrix}\lambda\mu & \mu^2\\ -\lambda^2 & -\lambda\mu\end{bmatrix}$ then A is

1) an idempotent matrix 2) nilpotent matrix
3) an orthogonal matrix 4) symmetric

Determinant and its properties:

28. $If\ A=\begin{bmatrix}a & b & c\\ b & c & a\\ c & a & b\end{bmatrix}$ then cofactor of a_{21} is

1) b^2 - ac 2) ac - b^2 3) a^2-bc 4) bc-a^2

29. $If\ A=\begin{bmatrix}1 & 4\\ 2 & 8\end{bmatrix} and\ B=\begin{bmatrix}x & y\\ y & x\end{bmatrix}$ then the cofactor of a_{21} in AB is

1) $-y-4x$ 2) $y+4x$ 3) $2x+8y$ 4) $-2x-8y$

30. $Det\begin{bmatrix}2 & 45 & 55\\ 1 & 29 & 32\\ 3 & 68 & 87\end{bmatrix}=.....$

1) 45 2) 64 3) 54 4) 32

31. det $\begin{bmatrix}1990 & 1991 & 1992\\ 1991 & 1992 & 1993\\ 1992 & 1993 & 1994\end{bmatrix}=$

1) 1992 2) 1993 3) 1994 4) 0

32. $\begin{vmatrix}(b+c)^2 & a^2 & b^2\\ b^2 & (c+a)^2 & b^2\\ c^2 & c^2 & (a+b)^2\end{vmatrix}$

$=\lambda abc(a+b+c)^3$ then $\lambda=....$

1) 0 2) 1 3) 2 4) 3

33. If $1,\omega,\omega^2$ are the cube roots of unity then

$\begin{vmatrix}1 & \omega & \omega^2\\ \omega & \omega^2 & 1\\ \omega^2 & 1 & \omega\end{vmatrix}=.....$

1) 0 2) 1 3) 2 4) -1

34. $\begin{vmatrix}a-b & p-q & x-y\\ b-c & q-r & y-z\\ c-a & r-p & z-x\end{vmatrix}=$

1) 0 2) 1 3) abc 4) xyz

35. $\begin{vmatrix}x+y+2z & x & y\\ z & y+z+2x & y\\ z & x & z+x+2y\end{vmatrix}=$

1) $(x+y+z)^3$ 2) $2(x+y+z)^3$
3) $x+y+z$ 4) $(x+y+z)^2$

36. If A,B,C are the angles of triangle ABC,

then $\begin{vmatrix}\sin 2A & \sin C & \sin B\\ \sin C & \sin 2B & \sin A\\ \sin B & \sin A & \sin 2C\end{vmatrix}=$

1) 1 2) 0 3) -1 4) $\frac{3\sqrt{3}}{8}$

37. Det $\begin{bmatrix}0 & p-q & p-r\\ q-p & 0 & q-r\\ r-p & r-q & 0\end{bmatrix}=$

1) (p-q) (q-r) (r-p) 2) 0
3) pqr 4) 4 pqr

38. $\begin{vmatrix}(x-2)^2 & (x-1)^2 & x^2\\ (x-1)^2 & x^2 & (x+1)^2\\ x^2 & (x+1)^2 & (x+2)^2\end{vmatrix}=$

1) 8 2) 16 3) -8 4) -16

39. If $\begin{pmatrix}1 & 2 & x\\ 4 & -1 & 7\\ 2 & 4 & -6\end{pmatrix}$ is a singular matrix,

then $x=$ **(EAM-2007)**

1) 0 2) 1 3) -3 4) 3

40. If $\begin{vmatrix}1+x & 2 & 3\\ 1 & 2+x & 3\\ 1 & 2 & 3+x\end{vmatrix}=0$ then $x=$

1) 1 2) -1 3) -6 4) 6

41. $\begin{vmatrix}1 & \log_b a\\ \log_a b & 1\end{vmatrix}=.....$

1) ab 2) b/a 3) a/b 4) 0

42. If $\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0$ then $x =$

1) $-1,-2$ 2) $1,2$ 3) $1,-2$ 4) $-1,2$

Adjoint of a Matrix :

43. If $P = \begin{bmatrix} 1 & 4 \\ 2 & 6 \end{bmatrix}$, then adj $(P) =$

1) $\begin{bmatrix} 1 & 4 \\ 2 & 6 \end{bmatrix}$ 2) $\begin{bmatrix} 6 & -4 \\ -2 & 1 \end{bmatrix}$

3) $\begin{bmatrix} 6 & -2 \\ -4 & 1 \end{bmatrix}$ 4) $\begin{bmatrix} 2 & 1 \\ 6 & 4 \end{bmatrix}$

44. If A_{3x3} and det A = 5 then det (adj A) =

1) 5 2) 25 3) 125 4) 1/5

45. If A is a square matrix such that

A $adj(A) = diag(k,k,k)$ then $|adjA| =$

1) k 2) k^2 3) k^3 4) k^4

Inverse of a Matrix:

46. The inverse of the matrix $\begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ is

(EAM-2008)

1) $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 4 & 3 \\ 3 & 3 & 4 \end{bmatrix}$ 2) $\begin{bmatrix} 1 & 3 & 1 \\ 4 & 3 & 8 \\ 3 & 4 & 1 \end{bmatrix}$

3) $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 3 & 4 \\ 3 & 4 & 3 \end{bmatrix}$ 4) $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$

47. If A$=\begin{bmatrix} 2 & 2 \\ -3 & 2 \end{bmatrix}$ B $=\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ then $(B^{-1}A^{-1})^{-1} =$

1) $\begin{bmatrix} 2 & 2 \\ -3 & 2 \end{bmatrix}$ 2) $\begin{bmatrix} 2 & -2 \\ 2 & 3 \end{bmatrix}$ 3) $\begin{bmatrix} 2 & 2 \\ -2 & 3 \end{bmatrix}$ 4) $\begin{bmatrix} -2 & 2 \\ 2 & 3 \end{bmatrix}$

48. The matrix A is such that A $\begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} -4 & 1 \\ 7 & 7 \end{bmatrix}$ then A =

1) $\begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$ 2) $\begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix}$ 3) $\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ 4) $\begin{bmatrix} -1 & 1 \\ 2 & 3 \end{bmatrix}$

49. If A is a 3×3 non singular matrix and $|adjA| = |A|^x$, $|adj(adjA)| = |A|^y$, $|A^{-1}| = |A|^z$, then the values of x, y, z in descending order.

1) x,y,z 2) z,y,x 3) z,x,y 4) y,x,z

50. If $A = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 8 \end{bmatrix}$, then $A^{-1} =$

1) $\begin{bmatrix} \frac{1}{9} & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$ 2) $\begin{bmatrix} 9 & 0 & 0 \\ 0 & \frac{1}{10} & 0 \\ 0 & 0 & 8 \end{bmatrix}$

3) $\begin{bmatrix} 9 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$ 4) $\begin{bmatrix} \frac{1}{9} & 0 & 0 \\ 0 & \frac{1}{10} & 0 \\ 0 & 0 & \frac{1}{8} \end{bmatrix}$

51. If the value of a third order determinant is 11, then the value of the determinant of $A^{-1} =$

1) 11 2) 121 3) 1/11 4) 1/121

52. If $A = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then $(AdjA)^{-1} =$

1) I 2) A 3) $-A$ 4) 0

Rank of a matrix:

53. The rank of the matrix $\begin{bmatrix} -1 & 2 & 5 \\ 2 & -4 & a-4 \\ 1 & -2 & a+1 \end{bmatrix}$ is

1)3 if $a=6$ 2)1 if $a=-6$

3)3 if $a=2$ 4) 2 if $a=-6$

54. The rank of $\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$ is :

1) 0 2) 1 3) 2 4) 3

55. The Rank of $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 4 & 6 & 2 \\ 1 & 2 & 3 & 2 \end{bmatrix}$ is

1) 1 2) 2 3) 0 4) 3

Solution of Simultaneous Equation non Homogeneous Linear Equations:

56. The number of solutions of the equations $2x-3y=5$, $x+2y=7$ is ....

1) 1 2) 2 3) 4 4) 0

57. If the system of equations $x+y+z=6$, $x+2y+\lambda z=0$, $x+2y+3z=10$ has no solution, then $\lambda=$

1) 2 2) 3 3) 4 4) 5

58. The equations $x+4y-2z=3$, $3x+y+5z=7$, $2x+3y+z=5$ have

1) A unique solution
2) Infinite number of solutions
3) No solution 4) Two solutions

59. If $x+y+z=1$, $ax+by+cz=k$, $a^2x+b^2y+c^2z=k^2$ has unique solution then $x=$.......

1) $\frac{(k-b)(c-k)}{(a-b)(c-a)}$ 2) $\frac{(k-c)(a-k)}{(b-c)(c-a)}$

3) $\frac{(k-a)(b-k)}{(b-c)(c-a)}$ 4) $(k-a)(k-b)(k-c)$

60. Solution of the system of equations

$\frac{2}{x}+\frac{3}{y}+\frac{10}{z}=4$, $\frac{4}{x}-\frac{6}{y}+\frac{5}{z}=1$,

$\frac{6}{x}+\frac{9}{y}-\frac{20}{z}=2, (x,y,z)=$

1) (1,2,3) 2) (2,3,5) 3) (3,2,5) 4) (5,3,2)

61. Consider the system of linear equations:

$x_1+2x_2+x_3=3$; $2x_1+3x_2+x_3=3$

$3x_1+5x_2+2x_3=1$. The system has **[AIE-2010]**

1) Infinite number of solutions
2) Exactly three solutions
3) A unique solution 4) No solution

62. If the system of equations $2x-3y+4z=0$, $5x-2y-z=0$, $21x-8y+az=0$ has infinity solutions then $a=$

1) -5 2) -4 3) 2 4) 4

63. The real value of 'a' for which the equations $ax+y+z=0$, $-x+ay+z=0$, $-x-y+az=0$ have non-zero solution is

1) 1 2) 0 3) -1 4) all the above

64. If the system of equations

$2x+3ky+(3k+4)z=0$

$x+(k+4)y+(4k+2)z=0$,

$x+2(k+4)y+(3k+4)z=0$

has non trivial solution then k=

1) -8 or $\frac{1}{2}$ 2) 8 or $-\frac{1}{2}$

3) -4 or $\frac{1}{2}$ 4) 4 or $-\frac{1}{2}$

LEVEL - II-KEY

01) 2	02) 3	03) 2	04) 4
05) 3	06) 2	07) 2	08) 2
09) 1	10) 2	11) 1	12) 4
13) 1	14) 1	15) 4	16) 4
17) 1	18) 3	19) 2	20) 1
21) 4	22) 4	23) 1	24) 3
25) 3	26) 1	27) 2	28) 2
29) 1	30) 3	31) 4	32) 3
33) 1	34) 1	35) 2	36) 2
37) 2	38) 3	39) 3	40) 3
41) 4	42) 1	43) 2	44) 2
45) 2	46) 4	47) 2	48) 3
49) 4	50) 4	51) 3	52) 2
53) 2	54) 4	55) 2	56) 1
57) 2	58) 3	59) 1	60) 2
61) 4	62) 1	63) 2	64) 1

LEVEL - II-HINTS

1. $a_{ij}=\frac{1}{2}(3i-2j) \Rightarrow a_{11}=\frac{1}{2}, a_{12}=\frac{-1}{2}$

and $a_{21}=2, a_{22}=1$

$\therefore A=\left[a_{ij}\right]_{2\times 2}=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$\therefore A=\begin{bmatrix} 1/2 & -1/2 \\ 2 & 1 \end{bmatrix}$

2. Equating corresponding elements

3. $2X=\begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix}$

4. Additive inverse of matrix A is - A

5. $X=\frac{-1}{2}(3A+5B)$

6. $[1\times 3]\times[3\times 3]\times[3\times 1]=1\times 1$

7. $A^2 = A.A$

8. $A=3I$, $A^4=3^4 I=81I=27A$

9. Using matrix multiplication

10. $AB=A\Rightarrow(AB)A=A^2$

$\Rightarrow A(BA)=A^2 \Rightarrow AB=A^2\Rightarrow A=A^2$

similarly $B=B^2$

11. $A^2+2A+1=(A+I)^2$

12. find aI, bE

13. $A^2=I \quad \therefore A^{2016}=(A^2)^{1013}=I$

14. $PQ=3\times 3$ matrix

15. find AB

16. $Tr[(2-i)A]=(2-i)Tr(A)$

$=(2-i)(2+i)=5$

17. $Tr(A-2B)=Tr(A)-2Tr(B)$

18. verify

19. put P = 2

20. put n = 3.

21. $O(A)=2\times 3, O(B)=3\times 2, O(C)=3\times 3$

$\Rightarrow O(B^T)=2\times 3\Rightarrow O(A+B^T)=2\times 3$

$C(A+B^T)$ is not defined as number of rows of $C\neq$ number of columns of $(A+B^T)$.

22. Conceptual

23. Find AA^T and A^TA we get $AA^T=A^TA=I$

24. Conceptual

25. $A^T=-A$

26. $p=\left(\frac{A+A^T}{2}\right)$

27. $A^2=0$

28. $(-1)\begin{vmatrix} b & c \\ a & b \end{vmatrix}$

29. find AB and then cofactors

30. applying $C_3\to C_3-C_2$

after that $C_2\to C_2-3C_3$ and expand

31. Use $R_2\to R_2-R_1, R_3\to R_3-R_1$

32. put $a=b=c=1$ find λ

33. Applying $C_1\to C_1+C_2+C_3$

34. $R_1\to R_1+R_2+R_3$

35. $x=y=z=1$

36. put $A=B=C=60^0$

37. Det of odd order skew-symmetric matrix is 0

38. put x = 0 and simplify

39. If A is singular matrix $|A|=0$

40. expand

41. expand

42. Substitute options and verify

43. $adjP=\begin{bmatrix}6&-4\\-2&1\end{bmatrix}$

44. $|adjA|=|A|^{n-1}=5^2=25$

45. $A(AdjA)=|A|I$

46. $A^{-1}=\dfrac{AdjA}{\det A}$

47. $(B^{-1}A^{-1})^{-1}=AB$

48. $AB=C\ \Rightarrow B=A^{-1}C$

49. $x=n-1, y=(n-1)^2, z=-1$

50. $A^{-1}=\dfrac{adjA}{|A|}$

51. $|A^{-1}|=\dfrac{1}{|A|}$

52. $(adjA)^{-1}=\dfrac{A}{|A|}$

53. C_1,C_2 are proportional there fore rank A<3

54. $\det A\neq 0$

55. every 3x3 submatrices of A is singular .

56. $\det A\neq 0$

57. $\det A=0$

58. Rank of A $\neq$ Rank AD

59. using Cramers rule $\therefore x=\dfrac{\Delta_1}{\Delta}$

60. Verification

61. Rank of A $\neq$ Rank AD

62. det A=0

63. $\begin{vmatrix}a&1&1\\-1&a&1\\-1&-1&a\end{vmatrix}=0\Rightarrow a=0$

64. det A =0 we get $2k^2+15k-8=0$

LEVEL - III

Algebra of Matrices:

1. **For the matrix** $A=\begin{bmatrix}3&2\\1&1\end{bmatrix}$**, the values of 'a' and 'b' such that** $A^2+aA+bI=O$ **are**

1) 2,3 2) 1,4 3) -4,1 4) -2,3

2. **If** $A=\begin{bmatrix}2&2&2\\2&2&2\\2&2&2\end{bmatrix}$ **then** $A^3-35A=$

1) A 2) 2A 3) 3A 4) 4A

3. $A=\begin{bmatrix}0&0&x\\0&x&0\\x&0&0\end{bmatrix}, A^{100}=$

1) $\begin{bmatrix}0&0&x^{100}\\0&x^{100}&0\\x^{100}&0&0\end{bmatrix}$ 2) $\begin{bmatrix}x^{100}&0&0\\0&x^{100}&0\\0&0&x^{100}\end{bmatrix}$

3) $\begin{bmatrix}o&x^{100}&o\\o&o&x^{100}\\x^{100}&o&o\end{bmatrix}$ 4) $\begin{bmatrix}o&x^{100}&o\\x^{100}&o&o\\o&o&x^{100}\end{bmatrix}$

4. **If A=** $\begin{pmatrix}0&0\\1&1\end{pmatrix}$ **then the value of** $A+A^2+A^3+......+A^n=$

1) A 2) nA 3) $(n+1)A$ 4) 0

5. **The number of** 2×2 **matrices that can be formed by using 1,2,3,4 when repetitions are allowed is**

1) 24 2) 12 3) 6 4) 256

Special types of Matrices, Symmetric & skew-Symmetric Matrices:

6. **If** $[3x^2+10xy+5y^2]=[x\ y]A\begin{bmatrix}x\\y\end{bmatrix}$, **and A is a symmetric matrix then A =**

1) $\begin{bmatrix}3&10\\10&5\end{bmatrix}$ 2) $\begin{bmatrix}10&3\\5&10\end{bmatrix}$

3) $\begin{bmatrix} +3 & -5 \\ -5 & +5 \end{bmatrix}$ 4) $\begin{bmatrix} 3 & 5 \\ 5 & 5 \end{bmatrix}$

7. **A square matrix A is said to be nilpotent of index m. If $A^m = 0$ now, if for this A,**

$(I-A)^n = I + A + A^2 + A^3 + ... + A^{m-1}$ **, then n is equal to**

1) 0 2) m 3) -m 4) -1

8. **Let A be the set of all 3×3 skew-symmetric matrices whose entries are either $-1, 0, or\ 1$ if there are exactly three 0's, three 1's then the number of such matrices is**

1) 3 2) 6 3) 8 4) 9

9. **The maximum number of different possible non-zero entries in a skew-symmetric matrix of order 'n' is**

1) $\frac{1}{2}(n^2-n)$ 2) $\frac{1}{2}(n^2+n)$

3) n^2 4) (n^2-n)

Determinants:

10. **If** $r^2 = a^2+b^2+c^2; s^2 = ab+bc+ca$ **then**

$$\begin{vmatrix} r^2 & s^2 & s^2 \\ s^2 & r^2 & s^2 \\ s^2 & s^2 & r^2 \end{vmatrix} =$$

1) $3abc - a^3-b^3-c^3$ 2) $a^3+b^3+c^3+3abc$

3) $(3abc-a^3-b^3-c^3)^2$ 4) 0

11. **If A=**$\begin{bmatrix} Cos\theta & Sin\theta \\ -Sin\theta & Cos\theta \end{bmatrix}$ **then** $\underset{n\to\infty}{Lt} \frac{1}{n}|A^n| =$

1) I 2) 0 3) A 4) $\frac{1}{n}A$

12. **If** $A_K = \begin{bmatrix} k & k-1 \\ k-1 & k \end{bmatrix}$ **then**

$|A_1| + |A_2| + ... + |A_{2015}| =$

1) 0 2) 2015 3) $(2015)^2$ 4) $(2015)^3$

13. **Matrix A is given by** $A = \begin{bmatrix} 6 & 11 \\ 2 & 4 \end{bmatrix}$ **then the determinant of** $A^{2015} - 6A^{2014}$ **is**

1) 2^{2016} 2) $(-11)2^{2015}$

3) $-2^{2015}\times 7$ 4) $(-9)2^{2014}$

14. **If** $\begin{vmatrix} x^n & x^{n+2} & x^{n+3} \\ y^n & y^{n+2} & y^{n+3} \\ z^n & z^{n+2} & z^{n+3} \end{vmatrix}$

$= (x-y)(y-z)(z-x)\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)$ **then value of n is**

1) -1 2) -2 3) 1 4) 2

15. **If** $\begin{vmatrix} 0 & \cos x & -\sin x \\ \sin x & 0 & \cos x \\ \cos x & \sin x & 0 \end{vmatrix}^2 = \begin{vmatrix} 1 & -a & a \\ -a & 1 & a \\ a & a & 1 \end{vmatrix}$ **then a =**

1) sin x 2) cos x

3) sin x .cos x 4) sinx-cos x

16 **If** $\begin{vmatrix} a & b & c \\ m & n & p \\ x & y & z \end{vmatrix}$ **= k, then** $\begin{vmatrix} 6a & 2b & 2c \\ 3m & n & p \\ 3x & y & z \end{vmatrix} =$

1) k/6 2) 2k 3) 3k 4) 6k

17. **If** $\begin{vmatrix} a+b & b+c & c+a \\ c+a & a+b & b+c \\ b+c & c+a & a+b \end{vmatrix} = t\times$ **det of circulant matrix whose elements of first column are** a, b, c **then** $'t'$ **equals**

1) 5 2) 6 3) -2 4) 2

18. **If** x, y, z **are integers in A.P lying between 1 and 9 and** $x15, y41$ **and** $z31$ **are three digit numbers,then the value of**

$$\begin{vmatrix} 5 & 4 & 3 \\ x51 & y41 & z31 \\ x & y & z \end{vmatrix}$$ **is:**

1) $x+y+z$ 2) $x-y+z$

3) 0 4) $x+2y+z$

19. **If the entries in a** 3×3 **determinant are either 0 or 1, then the greatest value of their determinants is**

1) 1 2) 2 3) 3 4) 9

20. **If** a,b,c **are the pth, qth, rth terms in H.P. then**

$$\begin{vmatrix} bc & p & 1 \\ ca & q & 1 \\ ab & r & 1 \end{vmatrix} =$$

1) 1 2) 0 3) abc 4) $a^2+b^2+c^2$

21. **If one of the roots of** $\begin{vmatrix} 3 & 5 & x \\ 7 & x & 7 \\ x & 5 & 3 \end{vmatrix} = 0$ **is -10, then the other roots are [EAM-2009]**

1) 3,7 2) 4,7 3) 3,9 4) 3,4

22. **The roots of** $\begin{vmatrix} x & a & b & 1 \\ \lambda & x & b & 1 \\ \lambda & \mu & x & 1 \\ \lambda & \mu & v & 1 \end{vmatrix} = 0$ **are independent of :**

1) λ,μ,v 2) a, b 3) λ,μ,v,a,b 4) 0,a

23. **If [.] denotes the greatest integer less than or equal to the real number under consideration, and** $-1 \le x < 0$; $0 \le y < 1$; $1 \le z < 2$, **then the value of the determinant**

$$\begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$$ **is**

1) $[x]$ 2) $[y]$ 3) $[z]$ 4) $[x]+[y]+[z]$

24. **If** α **is the repeated root of quadratic equation** $f(x)=0$ **and** $A(x), B(x)$ **and** $C(x)$ **are polynomials of degree 3,4 and 5 respectivelty, then**

$$\phi(x) = \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A^1(\alpha) & B^1(\alpha) & C^1(\alpha) \end{vmatrix}$$ **is divisible by**

1) $f(x)$ 2) $A(x)$ 3) $B(x)$ 4) $C(x)$

25. **If** $D_k = \begin{vmatrix} 1 & n & n \\ 2k & n^2+n+2 & n^2+n \\ 2k-1 & n^2 & n^2+n+2 \end{vmatrix}$ **and** $\sum_{k=1}^{n} D_k = 48$, **then** 'n' **equals**

1) 4 2) 6 3) 8 4) 10

26. **If** $a+b+c=0$ **and** $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$ **then** $x =$

1) 0 2) $\sqrt{\frac{3}{2}(a^2+b^2+c^2)}$

3) $-\sqrt{\frac{3}{2}(a^2+b^2+c^2)}$ 4) $0, \pm\sqrt{\frac{3}{2}(a^2+b^2+c^2)}$

27. **If** $a,b,c>0$ **and** $x,y,z \in R$ **then the determinant :**

$$\begin{vmatrix} (a^x+a^{-x})^2 & (a^x-a^{-x})^2 & 1 \\ (b^y+b^{-y})^2 & (b^y-b^{-y})^2 & 1 \\ (c^z+c^{-z})^2 & (c^z-c^{-z})^2 & 1 \end{vmatrix}$$ **is equal to :**

1) $a^x b^y c^z$ 2) $a^{-x}b^{-y}c^{-z}$ 3) 0 4) $a^{2x}b^{2y}c^{2z}$

28. **If** $A = \begin{vmatrix} 1+\cos\alpha & 1+\sin\alpha & 1 \\ 1+\cos\beta & 1+\sin\beta & 1 \\ 1 & 1 & 1 \end{vmatrix} \ne 0$, **then**

1) $\alpha = \beta$

2) $\alpha \ne \beta + n\pi$, n being any integer

3) $\alpha \ne \beta + \pi/2$ 4) $\alpha \ne \beta - \pi/2$

29. **If** ω **is a cube root of unity then**

$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ **has a factor**

1) $a+b\omega+c\omega^2$ 2) a-b-c
3) $a-b\omega^2-c\omega$ 4) a+b-c

30. **If** $a=\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}$ **then** $\begin{vmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{vmatrix}=$

1) purely real 2) purely imaginary
3) complex number 4) rational

31. $a\neq p, b\neq q, c\neq r$ **and** $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix}=0$ **then**

the value of $\frac{p}{p-a}+\frac{q}{q-b}+\frac{r}{r-c}=$

1) 1 2) 2 3) 3 4) c

32. **If** $\begin{vmatrix} p & q-y & r-z \\ p-x & q & r-z \\ p-x & q-y & r \end{vmatrix}=0$ **then the value**

of $\frac{p}{x}+\frac{q}{y}+\frac{r}{z}$ **is**

1) 0 2) 1 3) 2 4) $4pqr$

33. **If** $\Delta=\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ **and**

$\Delta_1=\begin{vmatrix} a_1+pb_1 & b_1+qc_1 & c_1+ra_1 \\ a_2+pb_2 & b_2+qc_2 & c_2+ra_2 \\ a_3+pb_3 & b_3+qc_3 & c_3+ra_3 \end{vmatrix}$ **then** $\Delta_1=$

1) Δ(1+pqr) 2) Δ(1+p+q+r)
3) Δ(1-pqr) 4) Δ

34. **If** $\begin{vmatrix} (1+x) & (1+x)^2 & (1+x)^3 \\ (1+x)^4 & (1+x)^5 & (1+x)^6 \\ (1+x)^7 & (1+x)^8 & (1+x)^9 \end{vmatrix}$

$=a_0+a_1x+a_2x^2+.......$, **then,** a_1 **is equal to**

1) 1 2) 2 3) 0 4) 3

35. **If y = cosx,** $y_n=\frac{d^n(\cos x)}{dx^n}$ **then** $\begin{vmatrix} y_4 & y_5 & y_6 \\ y_7 & y_8 & y_9 \\ y_{10} & y_{11} & y_{12} \end{vmatrix}=....$

1) 0 2) -cosx 3) cosx 4) sin x

36. **The value of the determinant**

$\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \sin\left(\theta+\frac{2\pi}{3}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & \sin\left(2\theta+\frac{4\pi}{3}\right) \\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right) \end{vmatrix}$ **is**

1) 0 2) $\sin\theta$
3) $\cos\theta$ 4) $\sin\theta+\cos\theta$

37. **If a, b, c are sides of a triangle**

and $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}=0$ **then**

1) Δ ABC is equilateral
2) Δ ABC is right angled isosceles
3) Δ ABC is isosceles
4) Δ ABC is a right angle

38. **With the usual notation in a**

$\Delta ABC, \begin{vmatrix} 1 & 1 & 1 \\ \sin A & \sin B & \sin C \\ \sin^2 A & \sin^2 B & \sin^2 C \end{vmatrix}=$

1) $\frac{1}{8R^3}(a-b)(b-c)(c-a)$

2) $8R^3$ 3) $(a-b)(b-c)(c-a)$

4) $\frac{1}{8R}(a-b)(a-c)(b-a)$

39. **If** $f(\theta)=\begin{vmatrix}\cos\theta & 1 & 0\\ 1 & 2\cos\theta & 1\\ 0 & 1 & 2\cos\theta\end{vmatrix}$ **then range of** $f(\theta)$ **is**

1) $[0,1]$ 2) $[-1,0]$ 3) $[-1,1]$ 4) $\left[0,\frac{1}{2}\right]$

Adjoint Matrix:

40. **If A and B are square matrices of same order and A is non -singular, then for a positive integer 'n',** $\left(A^{-1}BA\right)^n$ **is equal to**

1) $A^{-n}B^nA^n$ 2) $A^nB^nA^{-n}$

3) $A^{-1}B^nA$ 4) $n\left(A^{-1}BA\right)$

41. **If A is orthogonal matrix of order 3 then** $\det\left(adj2A\right)=$

1) 4 2) 16 3) 27 4) 64

42. **If** $P=\begin{bmatrix}1 & \alpha & 3\\ 1 & 3 & 3\\ 2 & 4 & 4\end{bmatrix}$ **is the adjoint of a** 3×3 **matrix A and** $|A|=4$**, then** α **is equal to**

1) 11 2) 5 3) 0 4) 4

[AIE-2013]

Inverse Matrix:

43. **Let** $A=\begin{bmatrix}1 & -1 & 1\\ 2 & 1 & -3\\ 1 & 1 & 1\end{bmatrix}$ **and** $10B=\begin{bmatrix}4 & 2 & 2\\ -5 & 0 & \alpha\\ 1 & -2 & 3\end{bmatrix}$.

If B is the inverse of A, then α **is :**

1) -2 2) -1 3) 2 4) 5

44. **A is an involutary matrix given by** $A=\begin{bmatrix}0 & 1 & -1\\ 4 & -3 & 4\\ 3 & -3 & 4\end{bmatrix}$ **then the inverse of** $\frac{A}{2}$ **will be**

1) 2A 2) $\frac{A^{-1}}{2}$ 3) $\frac{A}{2}$ 4) A^2

45. **If** $A=\begin{bmatrix}a & b\\ 0 & c\end{bmatrix}$ **then** $A^{-1}+(A-aI)(A-cI)=$

1) $\frac{1}{ac}\begin{bmatrix}a & b\\ 0 & -c\end{bmatrix}$ 2) $\frac{1}{ac}\begin{bmatrix}-a & b\\ 0 & c\end{bmatrix}$

3) $\frac{1}{ac}\begin{bmatrix}c & -b\\ 0 & a\end{bmatrix}$ 4) $\frac{1}{ac}\begin{bmatrix}c & b\\ 0 & a\end{bmatrix}$

46. **Let** $A_\alpha=\begin{bmatrix}\cos\alpha & -\sin\alpha & 0\\ \sin\alpha & \cos\alpha & 0\\ 0 & 0 & 1\end{bmatrix}$

I) $A_\alpha A_\beta=A_{\alpha+\beta}$ II) $A_\alpha B_\beta=A_{\alpha\beta}$

III) $(A_\alpha)^{-1}=-A_\alpha$ IV) $(A_\alpha)^{-1}=A_{-\alpha}$

Then which of the above statements is / are correct

1) only II and III 2) only II and IV

3) only I and III 4) only I and IV

47. **If the product of the matrix** $B=\begin{bmatrix}2 & 6 & 4\\ 1 & 0 & 1\\ -1 & 1 & -1\end{bmatrix}$ **with a matrix A has inverse** $C=\begin{bmatrix}-1 & 0 & 1\\ 1 & 1 & 3\\ 2 & 0 & 2\end{bmatrix}$ **then** $A^{-1}=$

1) $\begin{bmatrix}-3 & -5 & 5\\ 0 & 9 & 14\\ 2 & 2 & 6\end{bmatrix}$ 2) $\begin{bmatrix}-3 & 5 & 5\\ 0 & 0 & 9\\ 2 & 14 & 16\end{bmatrix}$

3) $\begin{bmatrix}-3 & -5 & -5\\ 0 & 9 & 2\\ 2 & 14 & 6\end{bmatrix}$ 4) $\begin{bmatrix}-3 & -3 & -5\\ 0 & 9 & 2\\ 2 & 14 & 16\end{bmatrix}$

Solutions of Simultaneous Equation:

48. **If A, B, C are the angles of a triangle, the system of equations,** $(\sin A)x+y+z=\cos A$

$x+(\sin B)y+z=\cos B$

$x+y+(\sin C)z=1-\cos C$ **has**

1) no solution 2) unique solution
3) infinitely many solutions
4) finitely many solutions

49. The number of values of k, for which the system equations $(k+1)x+8y=4k$,
$kx+(k+3)y=3k-1$ has no solution , is [AIE-2013]

1) 1 2) 2 3) 3 4) Infinite

50. The system of equations $x+y+z=6$, $x+2y+3z=10$, $x+2y+\lambda z=k$ is inconsistent if $\lambda=\ldots..,\ k\neq\ldots\ldots..$

1) 3,7 2) 3,10 3) 7,10 4) 10,3

51. The values of 'm' for which the system of equations $3x+my=m$ and $2x-5y=20$ has a solution satisfying the conditions $x>0$, $y>0$ are given by the set.

1) $\{m\backslash m<-13/2\}$ 2) $\{m\backslash m>17/2\}$

3) $\{m\backslash m<-13/2\, or\, m>17/2\}$

4) $\{m>30\, or\, m<-15/2\}$

52. If the system of equations

$(k+1)^3x+(k+2)^3y=(k+3)^3$,

$(k+1)x+(k+2)y=k+3, x+y=1$

is consistent, then the value of k is [EAM-2010]

1) 2 2) -2 3) -1 4) 1

53. Let λ and α be real. The set of all values of λ for which the system of linear equations

$\lambda x+(\sin\alpha)y+(\cos\alpha)z=0,$

$x+(\cos\alpha)y+(\sin\alpha)z=0$,

$-x+(\sin\alpha)y-(\cos\alpha)z=0$

has a non trival solution is

1) $\left[0,\sqrt{2}\right]$ 2) $\left[-\sqrt{2},0\right]$ 3) $\left[-\sqrt{2},\sqrt{2}\right]$ 4) $\left[1,\sqrt{2}\right]$

54. If the equations

$(b+c)x+(c+a)y+(a+b)z=0$;

$cx+ay+bz=0$; $ax+by+cz=0$

have non-zero solutions then a relation among a, b, c is

1) $a=b=c=0$ 2) $a=2b=3c$
3) $a=b=c(\neq 0)$ 4) $a+2b+3c=0$

55. Given that $a\alpha^2+2b\alpha+c\neq 0$ and that the system of equations [EAM-2012]

$(a\alpha+b)x+ay+bz=0$,

$(b\alpha+c)x+by+cz=0$

$(a\alpha+b)y+(b\alpha+c)z=0$ **has a non trivial solution, then a,b,c are in**

1) A.P 2) G.P 3) H.P 4) A.G.P

56. If the equations $a(y+z)=x, b(z+x)=y$, $c(x+y)=z$ have nontrivial solutions, then

$\frac{1}{1+a}+\frac{1}{1+b}+\frac{1}{1+c}=$

1) 1 2) 2 3) -1 4) -2

LEVEL - III -KEY

01) 3	02) 1	03) 2	04) 2	05) 4	
06) 4	07) 4	08) 3	09) 4	10) 3	
11) 2	12) 3	13) 2	14) 1	15) 3	
16) 4	17) 3	18) 3	19) 2	20) 2	
21) 1	22) 2	23) 3	24) 1	25) 1	
26) 4	27) 3	28) 2	29) 1	30) 2	
31) 2	32) 3	33) 1	34) 3	35) 1	
36) 1	37) 3	38) 1	39) 3	40) 3	
41) 4	42) 1	43) 4	44) 1	45) 3	
46) 4	47) 3	48) 2	49) 1	50) 2	
51) 4	52) 2	53) 3	54) 3	55) 2	56) 2

LEVEL - III-HINTS

1. $A^2=\begin{bmatrix}11 & 8\\ 4 & 3\end{bmatrix}$

2. $A^2=\begin{bmatrix}12 & 12 & 12\\ 12 & 12 & 12\\ 12 & 12 & 12\end{bmatrix}$, $A^3=\begin{bmatrix}72 & 72 & 72\\ 72 & 72 & 72\\ 72 & 72 & 72\end{bmatrix}$

3. $A=\begin{bmatrix}0&0&x\\0&x&0\\x&0&0\end{bmatrix}\Rightarrow A^{100}=\begin{bmatrix}x^{100}&0&0\\0&x^{100}&0\\0&0&x^{100}\end{bmatrix}$

4. $A^2=A^3=A^4=........=A$

5. $\begin{bmatrix}\underline{4}&\underline{4}\\\underline{4}&\underline{4}\end{bmatrix}$

6. Verification

7. Let $B=I+A+A^2+....A^{m-1}$

$\Rightarrow B(I-A)=(I+A+A^2+...+A^{m-1})$

$(I-A)=1-A^m=I$

$\Rightarrow B=(I-A)^{-1}\Rightarrow n=-1$

8. In a skew-symmetric matrix, and diagonal elements are zero.

$\begin{bmatrix}0&\times&\times\\\times&0&\times\\\times&\times&0\end{bmatrix}$ also $a_{ij}+a_{ji}=0$ Hence, number of matrices $=2\times2\times2=8$

9. Let n = 3

10. Put a = 0, b = 1, c = 2 and verify the options

11. det A = 1, $\underset{n\to\infty}{Lt}\frac{1}{n}=0$

12. $|A_k|=(2K-1)$

13. $|A^{2015}-6A^{2014}|=|A|^{2014}|A-6I|=$

$2^{2014}\begin{vmatrix}0&11\\2&-2\end{vmatrix}=(-22)2^{2014}=-11(2)^{2015}$

14. Order of determinant is
$n+n+2+n+3=3n+5$
Order of R.H.S = 1+1+1-1=2
$\Rightarrow 3n+5=2\Rightarrow 3n=-3\Rightarrow n=-1$

15. $LHS=\begin{vmatrix}0&\cos x&-\sin x\\\sin x&0&\cos x\\\cos x&\sin x&0\end{vmatrix}\begin{vmatrix}0&\sin x&\cos x\\\cos x&0&\sin x\\-\sin x&\cos x&0\end{vmatrix}\left(\because|A|=|A^T|\right)$

16. Take out '3' from C_1 and '2' from $R_1\Rightarrow$ Det $=6k$

17. We know that

$\begin{vmatrix}a+b&b+c&c+a\\c+a&a+b&b+c\\b+c&c+a&a+b\end{vmatrix}=-2\begin{vmatrix}a&b&c\\b&c&a\\c&a&b\end{vmatrix}$

$\Rightarrow t=-2$

18. Put x = 5, y = 4, z = 3

19. Keep least of given values in principal diagonal, highest of given values in other places.

$\Rightarrow\begin{vmatrix}0&1&1\\1&0&1\\1&1&0\end{vmatrix}=2$

20. $\frac{1}{a},\frac{1}{b},\frac{1}{c}$ are in A.P, $\frac{1}{a}=A+(P-1)D$

$\frac{1}{b}=A+(q-1)D,\ \frac{1}{c}=A+(r-1)D$

$\begin{vmatrix}bc&p&1\\ca&q&1\\ab&r&1\end{vmatrix}=\frac{1}{abc}\begin{vmatrix}\frac{1}{a}&p&1\\\frac{1}{b}&q&1\\\frac{1}{c}&r&1\end{vmatrix}=0$

21. $R_1\to R_1+R_2+R_3$ and take common 10+x

22. Operate
$R_1\to R_1-R_2, R_2\to R_2-R_3, R_3\to R_3-R_4$,

we get $\begin{vmatrix}x-\lambda&a-x&0&0\\0&x-\mu&b-x&0\\0&0&x-\nu&0\\\lambda&\mu&\nu&1\end{vmatrix}=0$ on

expanding we get $(x-\lambda)(x-\mu)(x-\nu)=0\Rightarrow$
Roots are independent of a, b.

23. [x] = - 1, [y] = 0 , [z] = 1

Det = $\begin{vmatrix}0&0&1\\-1&1&1\\-1&0&2\end{vmatrix}=1=[z]$

24. $f(x)=a(x-\alpha)^2 \quad a\neq 0$

$\phi(\alpha)=\begin{vmatrix}A(\alpha)&B(\alpha)&C(\alpha)\\A(\alpha)&B(\alpha)&C(\alpha)\\A^1(\alpha)&B^1(\alpha)&C^1(\alpha)\end{vmatrix}=0$

i.e α is a root of $\phi(x)$

$$\phi^1(\alpha) = \begin{vmatrix} A^1(\alpha) & B^1(\alpha) & C^1(\alpha) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A^1(\alpha) & B^1(\alpha) & C^1(\alpha) \end{vmatrix} +0+0=0$$

$$\therefore \phi^1(\alpha) = (x-\alpha)^2 \psi(x)$$

Where $\psi(x)$ is a polynomial of maximum degree 3

$\phi(x) = f(x)\,\psi(x)$

25. $\sum_{k=1}^{n} 1 = n$, $\sum_{k=1}^{n} 2k = n(n+1)$, $\sum_{k=1}^{n}(2k-1) = n^2$

Exp =

$$\begin{vmatrix} n & n & n \\ n(n+1) & n^2+n+2 & n^2+n \\ n^2 & n^2 & n^2+n+2 \end{vmatrix} = 48$$

By $C_2 \to C_2 - C_1$

$C_3 \to C_3 - C_1$, $n(2n+4) = 48 \Rightarrow n = 4$

26. $R_1 \to R_1 + R_2 + R_3$

$$\Delta = \begin{vmatrix} a+b+c-x & a+b+c-x & a+b+c-x \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$$

$$(a+b+c-x)\begin{vmatrix} 1 & 1 & 1 \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$$

$(a+b+c-x) = 0$ or

$ab+bc+ca-a^2-b^2-c^2+x^2=0$

$x = 0, \pm\sqrt{\frac{3}{2}\left(a^2+b^2+c^2\right)}$

27. Operate $C_1 \to C_1 - C_2$

$$\begin{vmatrix} 4 & \left(a^x-a^{-x}\right)^2 & 1 \\ 4 & \left(b^y-b^{-y}\right)^2 & 1 \\ 4 & \left(c^x-c^{-z}\right) & 1 \end{vmatrix} = 4\begin{vmatrix} 1 & \left(a^x-a^{-x}\right)^2 & 1 \\ 1 & \left(b^y-b^{-y}\right)^2 & 1 \\ 1 & \left(c^x-c^{-z}\right) & 1 \end{vmatrix} = 0$$

28. Expanding the det, we get

$\sin(\alpha-\beta) \neq 0$

$\alpha \neq n\pi + \beta; n \in Z$

29. $a=b=c$, $1+\omega+\omega^2=0$

$a+b\omega+c\omega^2=0$

since $\Delta = 0$

30. $a = w^2 \Rightarrow \Delta = \begin{vmatrix} 3 & 0 & 0 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{vmatrix} == 3(a^2-a)$

31. $R_1 - R_2; R_3 - R_1$

32. $R_1 - R_2; R_2 - R_3$ we get

$$\Delta = \begin{vmatrix} x & -y & 0 \\ 0 & y & -z \\ p-x & q-y & r \end{vmatrix} = 0$$

$$\Rightarrow pyz + qzx + rxy = 2xyz \Rightarrow \frac{p}{x}+\frac{q}{y}+\frac{r}{z} = 2$$

33. Split Δ_1

34. Differentiate both sides with respect to ‘x’ and put x = 0, we get $a_1=0$

35. Two rows are identical

36. Applying $R_2 \to R_2 + R_3$

$$\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ -\sin\theta & -\cos\theta & -\sin 2\theta \\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right) \end{vmatrix} = 0$$

37. $R_2 \to R_2 - R_1$

$$\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ 4a & 4b & 4c \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix}$$

$= 2(a-b)(b-c)(c-a)$

38. use sinerule that is a = 2R sinA, b = 2R sinB, c = 2R sinC.

39. $f(\theta) = \cos 3\theta \Rightarrow$ range of f is $[-1,1]$

40. $\left(A^{-1}BA\right)^2 = \left(A^{-1}BA\right)\left(A^{-1}BA\right) = A^{-1}B^2A$

41. $|adj(2A)| = 4^3|A|^{n-1} = 64(1) = 64$

42. $\left|P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}\right|$, $|Adj\ A| = |A|^2, |Adj\ A| = 16$

$1(12-12) - \alpha(4-6) + 3(4-6) = 16$

$2\alpha - 6 = 16, \alpha = 11$

43. AB = I

44. $A^2 = I \Rightarrow A = A^{-1}$ also $(KA)^{-1} = \frac{1}{K}A^{-1}$

Hence $\left(\frac{1}{2}A\right)^{-1} = 2A^{-1} = 2A$

45. $|A - \lambda I| = 0$

$\Rightarrow (\lambda - aI)(\lambda - cI) = 0$

$\Rightarrow (A - aI)(A - cI) = 0 \Rightarrow A^{-1} = \frac{adjA}{\det A}$

46. I and IV are true.

47. $BA = C^{-1}$, $A = B^{-1}C^{-1}$

$$A^{-1} = CB = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 1 & 3 \\ 2 & 0 & 2 \end{bmatrix}\begin{bmatrix} 2 & 6 & 4 \\ 1 & 0 & 1 \\ -1 & 1 & -1 \end{bmatrix}$$

48. Hint : $\Delta = \begin{vmatrix} \sin A & 1 & 1 \\ 1 & \sin B & 1 \\ 1 & 1 & \sin C \end{vmatrix}$ by

$C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$

Expanding along C_3 we get

Δ = SinA (1 - sinB) (1 - SinC) + (1 - sinA) (1 - sinB) + (1 - sinA) (1 - SinC)

Since A, B, C are angles of a triangle

0<SinA, SinB, SinC $\leq$ 1

Also at most one of SinA, SinB, SinC can be equal to 1 . Then atmost two terms of Δ can be zero and remaining +ve. Then $\Delta > 0$. Hence unique solution.

49. For no solution

$\frac{k+1}{k} = \frac{8}{k+3} \neq \frac{4k}{3k-1}$ ....(1)

$\Rightarrow (k+1)(k+3) - 8k = 0$

or $k^2 - 4k + 3 = 0 \Rightarrow k = 1, 3$

But for k= 1, equation (1) is not satisfied Hence k=3

50 $[A.B] = \begin{bmatrix} 1 & 1 & 1:6 \\ 1 & 2 & 3:10 \\ 1 & 2 & \lambda : k \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1:6 \\ 0 & 1 & 2:4 \\ 0 & 0 & \lambda - 3 : k - 10 \end{bmatrix}$

$R_2 \to R_2 - R_1, R_3 \to R_3 - R_2$

Given System is in consistent

rank$(A) \neq rank(B)$. Hence $K \neq 10, \lambda = 3$

51.

,

If and system of equations is in consistent

By cramer's rule

or

52.

53. det A=0

Range of is

54.

55.

56.

gives

gives

Expandign by

LEVEL - IV

Algebra of Matrices:

1. and A

1) 0 2) 1 3) 1/2 4) -2

2. If

where A,B,C are matrices then B + C =

1) 2) 3) 4)

3. If and then

1) xy = 0 2) 3) xy = -1 4) xy =1

4. If and

are two matrices such that the product AB is the null matrix then

1) 0 2) multiple of

3) an odd multiple of 4) can not be determind

5. If A,B are two square matrices such that AB=B; BA=A and then =

1) 2)

3) 4)

6. The number of matrices that can be formed by using without repetition is

1) 24 2) 12 3) 6 4) 256

Special types of Matrices, Symmetric & Skew Symmertic Matrices:

7. If = A+B where A is symmetric matrix and B is skew - symmetric then A - B =

1) 2)

3) 4)

8. If is an nilpotent matrix of index '2' then k=

1) 2 2) -2 3) 3 4) -3

9. If A is 3x3 non-singular matrix that and , then is equal to (AIEEE-2014)

1) 2) I 3) 4)

Determinants:

10. If A and B are square matrices of order 3 such that then

1) -9 2) -81 3) -27 4) 81

11. If

Then

1) 1 2) 2 3) 3 4) 1/2

12.

1) $n(a+b+c)^2$ 2) $n(a+b+c)^3$
3) $n^3(a+b+c)$ 4) $n^2(a+b+c)^2$

13. Let = ,

then the value of **is equal to**

1) 0 2) -16 3) 16 4) -11

14. If A, B, C are the angles of a **then**

1) 2)
3) 4) 0

15.

1) abcd 2) 0 3) a+b+c+d 4) 1

16. If **are all positive and are the** **and** **terms of a geometric progression respectively, then the value of the determinant** **[EAM-2009]**

1) log xyz 2) (p-1)(q-1)(r-1)
3) pqr 4) 0

17. If f (x) = tan x and A, B, C are the angles of **then**

1) 0 2) -2 3) 2 4) 1

18. If

and **, then** **is equal to**

1) 2)
3) 4)

19. Let = **where**

a=1, b= **and c=** ^{2} **then** =

1) 2) - ^{2} 3) 1 4) -1

20. If the determinant

is expressible as **, then the value of m is**

1) -1 2) 0 3) 1 4) 2

21. Let t be a positive integer and

then the value of is equal to

1) 2) 0 3) 4)

22. If then

1) 2900 2) 2800 3) 2700 4) 2600

23. The value of the determinant

is

1) independent of 2) Independent of

3) independent of 4) cannot be said

24. then

1) 2) 3) 4)

25. The number of distinct real roots of

in the interval

is

1) 0 2) 2 3) 1 4) 3

26. The sum of two non integral roots of

is

1) 4 2) -4 3) -25 4) 25

27. If and is not a root of ,then

1) are in A.P 2) are in G.P

3) are in H.P 4) are in A.G..P

28. If , then

1) 0 2) 2 3) 4)

29. Let

then equals

1) 2) 3) 4)

30. If and are 3 - digit even natural numbers and ,

then is

1) divisible by 2 but not necessarily by 4

2) divisible by 4 but not necessarily by 8

3) divisible by 8 4) divisible by 25

31. If the value of is

1) -1 2) 0 3) 1 4) a^6

Adjoint Matrix:

32. A, B, C are cofactors of elements, a, b, c in then the value of (2A+4B+7C) is equal to

1) 0 2) 2 3) -1 4) 4

33. If then is

1) 2) 3) 4)

34. If ,then

1) 2)

3) 4)

35. , ,

then

1) 2)

3) 4)

36.

1) 2)

3) 4)

37. (EAM-2011)

1) 2) 3) 4) -

38. A nontrivial solution of the system of equations is given by =

1) 2)

3) 4)

39. The system of equations , , has no solution, if is [AIE-2005]

1)1 2)not-2 3)either-2 or 1 4)-2

40. If ' ' is cube root of unity and x + y + z = a x + y + = b,x + + =c then which of the following is correct

1) 2)

3) 4) All the above

41. If the trivial solution is the only solution of the system of equations , , . Then the set of all values of k is

1) 2)

3) 4)

42. **If the system of equations ,**
;
has a non trivial solution then

1)1 2)1- 3)2 4)-2

43. **The system of equations**
; has
non trivial solution then

1)2 2)-2 3)0 4)1

44. **If the system of linear equations**
; ,
has non trivial solution then

1)6 212 3)18 4)16

45. **The number of values of k for which the linear equations ,**
, posses a non zero solution is:

1)3 2)2 3)1 4) zero

LEVEL - IV-KEY

01) 1	02) 1	03) 1	04) 3
05) 2	06) 1	07) 1	08) 2
09) 2	10) 2	11) 2	12) 2
13) 4	14) 4	15) 2	16) 4
17) 3	18) 1	19) 4	20) 2
21) 2	22) 1	23) 2	24) 2
25) 3	26) 2	27) 2	28) 2
29) 2	30) 1	31) 2	32) 1
33) 1	34) 1	35) 1	36) 2
37) 1	38) 4	39) 4	40) 4
41) 3	42) 1	43) 1	44) 1
45) 2			

LEVEL - IV-HINTS

1. $A^2 =$ I equating 1st row x 1st column elements on both sides
2.
3. A=0 (or)
4. AB=0
5. Put n = 2 and verify
6. Rq no of ways = 4 x 3 x 2 x 1
7.
8.
9.
10.
11.

=

12. Put n = 2, a = 1, b =1, c= 2
13. expand and compare
14. Expand along 1^{st} column
15. Put a = 1, b =1, c = 2, d = 2
16.

and substitute

17. $$\begin{vmatrix} TanA & 1 & 1 \\ 1 & TanB & 1 \\ 1 & 1 & TanC \end{vmatrix}$$

18. Taking common from and applying and . We obtain

19.

20. For the first determinant write C_1 $C_1 + C_2 + C_3$

21. Applying the property summation of determinant to column and the L.H.S

Now i.e., are identical

22.

23.

24.

=

25. By R_1 $R_1 + R_2 + R_3$

(sinx+2cosx)

Now C_3 $C_3 - C_1$, C_2 $C_2 - C_1$

$(\text{Sinx} + 2\cos x)(\sin x - \cos x)^2 = 0$

$\tan x = 1$ $x =$

26.

the two non integral roots are roots of the equation

sum of roots

27.

28. Expand

29. Applying $C_1 \to C_1 - (Sinx)C_3$

$C_2 \to C_2 + (Cosx)C_3$

$$\Delta(x) = \begin{vmatrix} 1 & 0 & -Sinx \\ 0 & 1 & Cosx \\ Sinx & -Cosx & 0 \end{vmatrix}$$

applying $R_3 \to R_3 - SinxR_1 + CosxR_2$

$$\Delta(x) = \begin{vmatrix} 1 & 0 & -Sinx \\ 0 & 1 & Cosx \\ 0 & 0 & 1 \end{vmatrix} = 1$$

30. As are natural numers, each of is divisible by 2.

Let for . Thus

where is some natural number. Thus, is divisible by 2. That may not be divisible by 4 can be seen by taking the three numbers as 112, 122 and 134. Note that which is divisible by 2 but not by 4.

31. Differentiate 1st row n times

32. =-13

33. We know that

, if ; provided

order of A is n

But

34.

35.

Find SAS^{-1}

36.

37. Use characterstic equation

38.

taking the system becomes

,

39. det A=0,then we get or -2

but the equation has infinitely many solutions.

40. Solve 1,2,3 by General method.

41.

42. det A=0

43. det A=0

44. det A=0

45.

LEVEL - V

1. If and

then

+

1) 6 2) 9 3) 12 4) 15

2. If (where) satisfies the equation ,then

1) 2)

3) 4) all the above

3. If then

1)1 2)0 3)A 4)

4. **Matrix** , **if** **and**

then **is equal to**

1) 2)

3) 4)

5. **If A and B are different matrices satisfying** **and** **,then (AIE-2012)**

1) must be zero

2) must be zero

3) as well as det(A-B) must be zero.

4) At least one of det or det (A-B) must be zero.

6. **If A, B and C are** **matrices and** **, then the value of the** **is equal to**

1) 2) 3) 4)

7. **If**

1) 1 2) -1 3) 0 4)

8. **The maximum and minimumvalues of** **determinant whose elements belong to {0,1,2,3}is**

1) 2) 3) 4)

9. **Let**

then det A is equal to

1) 2)

3) 4)

10. **Let**

then

1) 2)

3) 4)

11. **If** **and**

If **,then**

1) 2)

3) 4)

12. **Let**

then the value of

where **and** **,is**

1) -2 2) 0 3) -2b 4) 2b

13. **If** ,

and and ,then the values of x and y are

1) 2)

3) 4)

14. The value of the determinant

is

1) 0 2) dependent only on

3) dependent only on

4) dependent on

15. Let be the set of all third order determinants that can be formed with the distinct non-zero real numbers then

1) 2)

3) for all i, j 4) None of the above

16. If

then equals

1)

2)

3)

4) 0

17. If

then B is equal to

1) 0 2) 1 3) 2 4) 4

18. If then

equals

1) 1 2) -1 3) 2a 4) 0

19. If then

is equal

1) 3 2) 5 3) 7 4) 0

20. If for , then

D is: (AIE-2007)

1) divisible by neither nor

2) divisible by both and

3) divisible by but not

4) divisible by but not

21. Let and

, where are in A.P. then the common difference

1) 1 2) 2 3) 3 4) 4

22.

1) 2)

3) 4) 0

23. D_r = D_r =

1) nr 2) 0 3) 4) $2n-n^2$

24. **If and**

=

,then is equal to (AIE-2014)

1) 1 2) -1 3) 4)

25. **The value of the determinant**

where ,is

1) 2)

3) 4)

26. **Let a, b, c be such that b(a + c) 0. If**

then the value of n is (AIE-2009)

1) zero 2) any even integer

3) any odd integer 4) any integer

27. **Let A be a square matrix all of whose entries are integers. Then, which one of the following is true? (AIE-2008)**

1) If det (A) = 1, then A^{-1} exists but all its entries are not necessarily integers

2) If det (A) 1, then A^{-1} exists and all its entries are non-integers

3) If det (A) 1, then A^{-1} exists and all its entries are integers

4) If det (A) = 1, then A^{-1} need not exist

28. A triangle has vertices A_i**(x_i,y_i) for i = 1, 2,3. Then**

means

1) medians of the triangle A_1,A_2,A_3 are concurrent;

2) the triangle A_1,A_2,A_3 is right angled at A_3;

3) the triangle A_1,A_2,A_3 is an equilateral triangle;

4) altitudes of the triangle A_1,A_2,A_3 are concurrent.

29. **For a fixed poitive integer , let**

then is equal to

1) - 8 2) - 16 3) - 32 4). - 64

30. **If , and are three polynomials of degree '2' and**

, then is

1) a one degree polynomial

2) a three degree polynomial

3) a two degree polynomial 4) a constant

31. **If**

then

1) 0 2) -1 3) -2 4) 2

32. **If** ,

then **is equal to**

1) 3 2) 6 3) 9 4) any real number

33. **The number of non singular matrices with four entries as 1 and all other entries as 0, is** **[AIE-2010]**

1) Lessthan 4 2) 5 3) 6 4) atleast 7

34. **If** , **and**

and **then** **is**

1)

2)

3)

4)

35. **If** **and**

then

1)

2)

3)

4)

36. **If**

then

1)

2)

3)

4)

37. **Let** **if** **are column matrices such that** **and**

then **equal to** **[AIE-2012]**

1) 2) 3) 4)

38. **If** **then**

1)

2)

3)

4) None of these

39. **If** **are non zero real numbers and if the equations**

has a non trivial solution then

equals

1) a+b+c 2) abc 3)1 4) a+b-c

40. **Let** **be positive numbers.The following system of equations in**

,

has

1) no solution 2) unique solution

3) infinitely many solutions

4) finitely many solutions

LEVEL - V-KEY

01) 1	02) 4	03) 2	04) 3
05) 4	06) 2	07) 3	08) 3
09) 2	10) 4	11) 3	12) 1
13) 4	14) 4	15) 2	16) 4
17) 1	18) 4	19) 1	20) 2
21) 2	22) 1	23) 2	24) 1
25) 2	26) 3	27) 3	28) 4
29) 4	30) 4	31) 4	32) 4
33) 4	34) 1	35) 1	36) 2
37) 4	38) 1	39) 2	40) 4

LEVEL - V-HINTS

1.

=

2. We have,

As, A satisfies,

and

,

3. The roots of the characteristic equation are 1,2,3

there fore 0

4.

5. and

on substracting equations (i) and (ii) we set

6.

7.

8. Keep least of given values in principal diagonal, highest of given values in other places.

9. Multiply by z and by y and and applying

or put x=0,y=1,z=2 and verify the options

10. By putting x=0 on both sides of the equation we have differentiating both sides and then putting we get

11. Consider the ,using

Using

using

12. Using

13. Given that, and

and

Using cramer's rule, we have

14. The given determinant can be rewritten as

15. Total no. of third order determinants with distinct non-zero real numbers as elements is 9!. These determinants can be grouped into two groups each containing determinansts such that correpsonding to each determinant in a group there is another determinant in the other group which is obtained by interchanging two consecutive rows of the determinant in the first group.

Sum of the values of the determinants is 0.

16. The given determinant can be rewritten as

17. Differentiating w.r.t 'x' on both sides

Put x = 0

;

B = 0.

18. then

assume

,

is an odd function

.

19. Put and verify

20.

on applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$

21. Let a, b, c, d are in A.P. with common difference 'k' operate $R_3 \to R_3 - R_1$, $R_2 \to R_2 - R_1$

$$\Delta(x) = \begin{vmatrix} x+a & x+b & x+a-c \\ k & k & 2k-1 \\ 2k & 2k & 4k \end{vmatrix}$$

Oprate $C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$

$$\Delta(x) = \begin{vmatrix} x+a & k & -c \\ k & 0 & k-1 \\ 2k & 0 & 2k \end{vmatrix} = -2k^2$$

$$\int_0^2 \Delta(x)\,dx = -16 \Rightarrow k = 2$$

22. Taking from

23.

taking common from

then and are proportional

24. Let

applying and

then

Hence, k=1

25. By expanding

,

26.

=0

27. As det (A) = 1, A^{-1} exists and

All entries in adj (A) are integers,

has integer entries.

28. If altitudes of a triangle are concurrent at H(0, 0), then AH $\perp A_2A_3$, $\frac{y_1-0}{x_1-0}\cdot\frac{y_2-y_3}{x_2-x_3}=-1$

$\Rightarrow y_1(y_2-y_3)+x_1(x_2-x_3)=0 \quad \Rightarrow \Delta=0$

29. Taking (n-1)! common from !

from and !from ,we get

Where

30. .

31. Defferentiating and put $x=0$

32. expand $f(x)$ and then integrate

33. Consider .by placing 1 in any one of the 6 position and 0 elsewhere.
we get 6 non-singular matrices.

similarly, given atleast one non-singular.

34. Here

,

35. Characteristic equation of A is

,

36. , i.e

,

37. , But

38. ,

39. Rewritten the given equations are

,

has a non trivial solution then

40. Adding 3 equations we get

Substracting first,second,third equation.from this we get

LEVEL - VI

These questions have Statement-I & statement-II. Each question has four choices (1),(2),(3),(4) out of which only one is correct. Choose the correct choice as per following.

1) Statement-I & statement-II both are true & statement-II is the correct explanation of Statement-1.

2) statement-I and statement-II both are ture & Statement-II is not explain statement-I.

3) Statement-I is true but Statement-II is false.

4) Statement-I is false but Statement-II is true.

1. **Statement-I: Rank of matrix is defined for only square matrices**
Statement-II: Trace of matrices of a matrix is defined for square matrices only.

2. **If is a square matrix such that**

List -I	List-II
A)	1. Symmetric matrix
B)	2. Skew symmetric matrix
C)	3. Unit matrix
D)	4. Diagonal matrix
	5. Upper triangular

Correct match of List -I from List-II

	A	B	C	D
1)	3	4	5	1
2)	3	4	5	2
3)	3	5	4	2
4)	1	5	3	2

3. **Statement-I:** Let A be a square matrix given by , P is symmetric matrix, Q is skew symmetric matric then Q is given by

Statement-II: For every square matrix A, is skew symmetric.

4. Let then

Statement-I: =

Statement-II:

5. **Statement-I:** The system of non-homogeneous equations , , has no solution.

Statement-II: detA=0 and where

6. **Statement-I:** Let A be an unknown matrix of order and be an identitiy matrix of order ,then ,is satisfied by more than one matrix of order .

Statement-II: All matrices of the form are involutary matrices for all real values of such that .

7. **Statement-I:** If A is matrix of order ,then

(Where represents determinant of A)

Statement-II: If A is a matrix of order , then .

8. **Statement-I:** The determinant of a matrix is 0 where for i and j .

Statement-II: The determinant of a skew-symmetric matrix of odd order is zero.

9. **Statement-I:** If then the value of is

Statement-II: If

where (i=1,2,3) are polynomials in x.

10. **Statement-I:** The systemof the equations has no solution

Statement-II: The system of equations has no solution, if

11. **Statement-I:** is neither symmetric nor skew symmetric

Statement-II: The matrix A cannot be expressed as a sum of symmetric and anti-symmetric matrices.

12. **If A is a matrix of order 2 x 2**

Statement-I: *adj (adjA) = A*

Statement-II: . **(AIE-2009)**

13. **Statement-I:** If are in G.P (for all i) then

Statement-II: The determinant of matrix is zero when elements of each row are in A.P.

14. **Statement-I:** If a, b, c are distinct and x,y,z are not all zero given that , , then

Statement-II: if a,b,c are distinct.

15. **Statement-I:** There are only finitely many matrices which commute with the matrix

Statement-II: If A is non-singular, then it commutes with I, adj A and

16. **Statement-I**: If the system of equations , , has nonzero solutions,then

Statement-II: If the system of equations AX=0 has a nonzero solution then A is singular.

17. Let be the roots of the equation ; a, b, .

Statement-I:

Statement-II: Any cubic equation has at least one real root.

18. **Statement I: If A is square matrix such that then**

Statement II: Let A= and

then

19. ( )

Statement-I: Factors of are

Statement-II: Taking the value of n=1 then the value of the above determinant

(abc) (a-b) (b-c) (c-a)

20. **Consider the system of equations**

,

Statement-I: The system of equations has no solution for

Statement-II: The determinant

, for

21. **Statement-I: If A is an matrix, then ,where m is any scalar**

Statement-II: If U is a matrix obtained of a matrix V by multiplying any row or column by a scalar m, then

22. **Let A and B be two symmetric matrices of order 3.**

Statement - 1: A(BA) and (AB)A are symmetric matrices.

Statement - II: AB is symmetric matrix if matrix multiplication of A and B is commutative. [AIE-2011]

23. **Let A be 2×2 matrix with real entries. Let I be the 2×2 identity matrix. denote by tr(A), the sum of diagonal entries of A, assume that $A^2 = I$. [AIE-2010]**

Statement I: Tr (A)=0

Statement II: |A|=1

24. **The entries in a 3×3 determinants are either 1 or -1,then match the following.**

i) Total number of such determinants	**a) 4**
ii)The number of such determinants whose values are 6	**b) -3**
iii)The maximum value of such a determinant	**c) 512**
iv) The minimum value of trace of such determinants.	**d) 0**

1) (i) ii) iii) iv)

2) (i) ii) iii) iv)

3) (i) ii) iii) iv)

4) i) ii) iii) iv)

25. **Let A be matrix with real entries. Let I be the identity matrix. Denote by tr(A), the sum of diagonal entries of A .Assume that . [AIE-2008]**

Statement I: If and then det(A)=-1

Statement II: If and then

LEVEL - VI-KEY

01) 2	02) 2	03) 2	04) 1	05) 3
06) 1	07) 3	08) 1	09) 4	10) 3
11) 3	12) 1	13) 1	14) 4	15) 4
16) 1	17) 2	18) 2	19) 1	20) 1
21) 1	22) 2	23) 3	24) 3	25) 3

LEVEL - VI-HINTS

1. Rank of matrix and trace of matrix is defined for only square matrices

2. Conceptual

3. Let

is skew symmetric.

Now

which is skew symmetric matrix.

Statement-1 & statement-2 both are individually true but statement-2 is not the correct explanation of the Statement-1.

4. and

,

Statement-2 is ture

Now =

Statement-1 and Statement-2 are individually ture and Statement-1 is correct explanation of Statement-1.

5. The given system can be written as AX=B, where

Here,

and

So, the statement-1 is true but statement-2 is false.

,

The given statement is inconsistent i.e., is has no solution)

6.

Hence, statements l and ll are correct, statement ll is the correct explanation of statement l.

7. Statement -2 is false because correct formula is

where n is orde of square matrix A.

8. key-1

9. Statement -1 is false but Statement-2 is true.

10. Statement-1 is true Statement-2 is false.
The system of equations

has to solution if .

11. Statement-1 is ture but Statement-2 is false

12. conceptual

13. , where r is the common ratio.

Applying

14. Let

Now the other bracket cannot vanish in the right of the correct statement-2.

The system can have non-zero solutions (x,y,z are not all zero)

If

statement-1 is false.

15. The statement-2 is true since

But a matrix can commute with general order matrices which may be infinite in number

Let be a matrix which commute with A, them .

The above four relations are equivalent to only two independent relations.

If ,then

Thus, are all possible matrices which commute with the given matrix

and c being any arbitrary complex numbers, thus, assertion A is false.

16. 1) Statement-1 is true.

17. 2) Statement-1 is true, because

,as

$\alpha+\beta+\gamma=0$

Statement-2 is also true, because any polynomial equation over reals has imaginary roots in pair and hence any cubic equation has at least one real root. However, statement-2 is not a correct reasoning of statement-1

Hence, (2) is the correct answer.

18. Statement I:Expand

StatementII: $A^K=0\forall K\geq 2$

$(A+I)^{50}=I+50A$

$(A+I)^{50}-50A=I\Rightarrow a=1,b=0,c=0,d=1$

19. By keeping a = b R_1,R_2 are similar, etc

20. (a) In the usual notation

$$D=\begin{vmatrix}1 & -2 & 3\\ -1 & 1 & -2\\ 1 & -3 & 4\end{vmatrix}=0$$

$$D_1=\begin{vmatrix}-1 & -2 & 3\\ k & 1 & -2\\ 1 & -3 & 4\end{vmatrix}=3-k$$

$$D_2=\begin{vmatrix}1 & -1 & 3\\ -1 & k & -2\\ 1 & 1 & 4\end{vmatrix}=k-3$$

$$D_3=\begin{vmatrix}1 & -2 & -1\\ -1 & 1 & k\\ 1 & -3 & 1\end{vmatrix}=k-3$$

$D_1,D_2,D_3=0$ if k = 3

Thus the system of equation has no solution if k is different from 3, i.e. $k\neq 3$ both the statement (1) and (2) are true and statement -2 also explains statement -1. The answer then is (a).

21. Conceptual

22. Given, $A^T=A,B^T=B$

now $(A(BA))^T=(BA)^T A^T=(A^T B^T)A^T=$

$(AB)A=A(BA)$ Similarly $(AB)A^T=(AB)A$

So, A(BA) adn (AB)A are symmetric matrics,

Again $(AB)^T=B^T A^T=BA$

Now if BA = AB, then AB is symmetric matrix

23. Let $A=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$

$$\begin{bmatrix}a & b\\ c & d\end{bmatrix}^2=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\quad(\because A^2=I)$$

$$\Rightarrow\begin{bmatrix}a^2+bc & ab+bd\\ ac+cd & bc+d^2\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

$\Rightarrow b(a+d)=0,c(a+d)=0$

and $a^2+bc=1,bc+d^2=1$

$\Rightarrow a=1,d=-1,b=c=0$

there fore $A=\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\Rightarrow\det A=-1$

Statement II: tr(A)=1-1=0

24. (i)$\rightarrow(c)$

Since every entry can take 2 values to-tal possible determinant $S=2^9$ =512

The value six will be attained only when three terms are 1 and negative term-1

$(\Delta=a_1b_2c_3+a_2b_3c_1+a_3b_1c_2-a_1b_3c_2-a_3b_2c_1)$

$-a_3b_2c_1)$

Which is impossible since if this is the case the multiplication of all results-1or 1

iii) $\rightarrow(a)$ iv) $\rightarrow(b)$

The maximum value of such a determinant

is 4 attained by $\begin{vmatrix}-1 & 1 & 1\\ 1 & -1 & 1\\ 1 & 1 & -1\end{vmatrix}$.

The matrix trace is obviously -3.

25. Conceptual